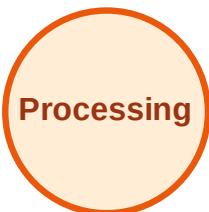


# BFS Traversal

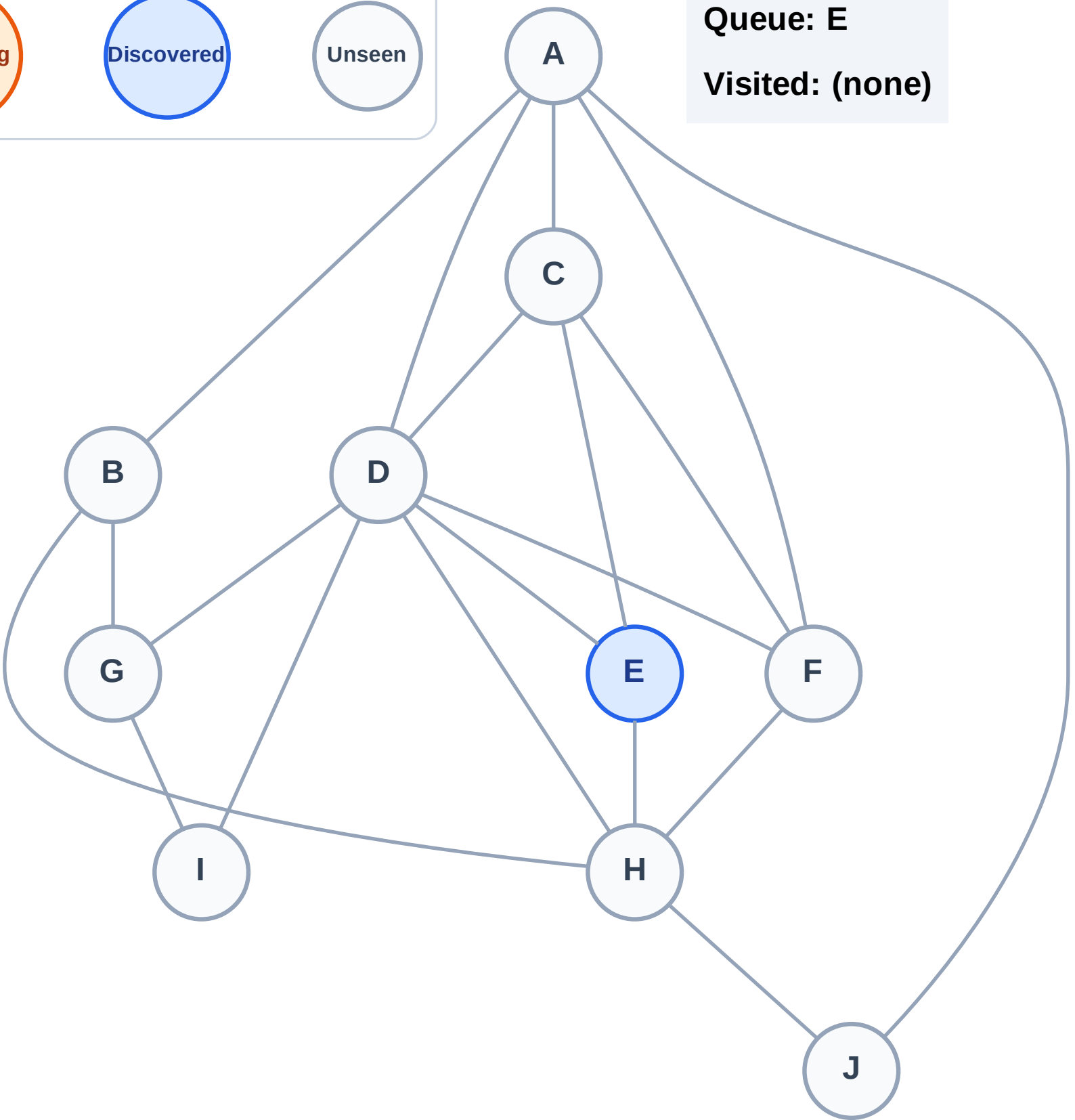
Step 1: Start at E

## Legend



Queue: E

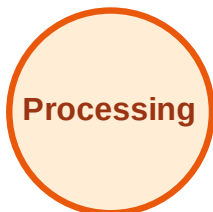
Visited: (none)



# BFS Traversal

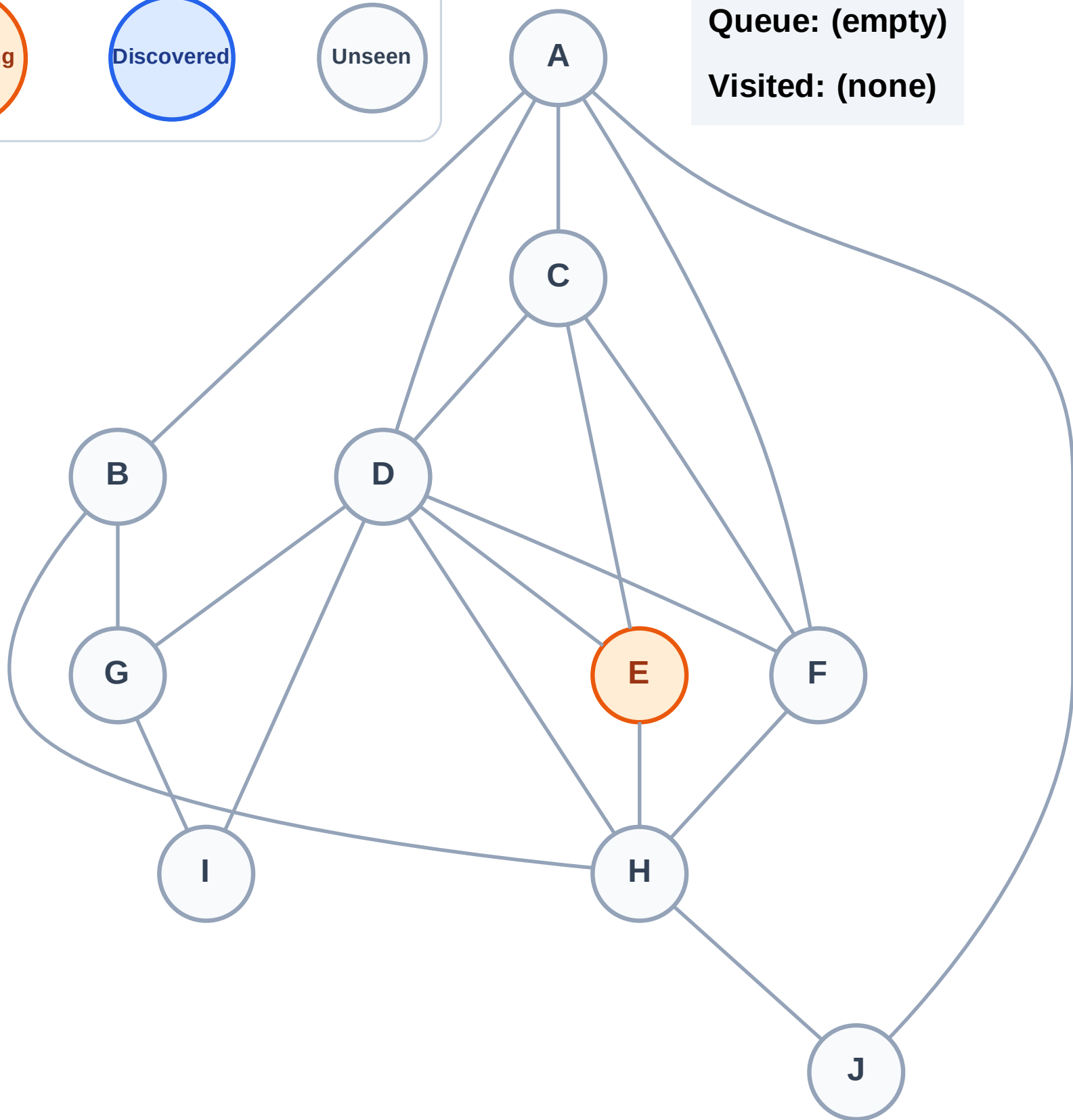
Step 2: Process node E

## Legend



Queue: (empty)

Visited: (none)



# BFS Traversal

Step 3: Discover C, D, H from E

Legend

Complete

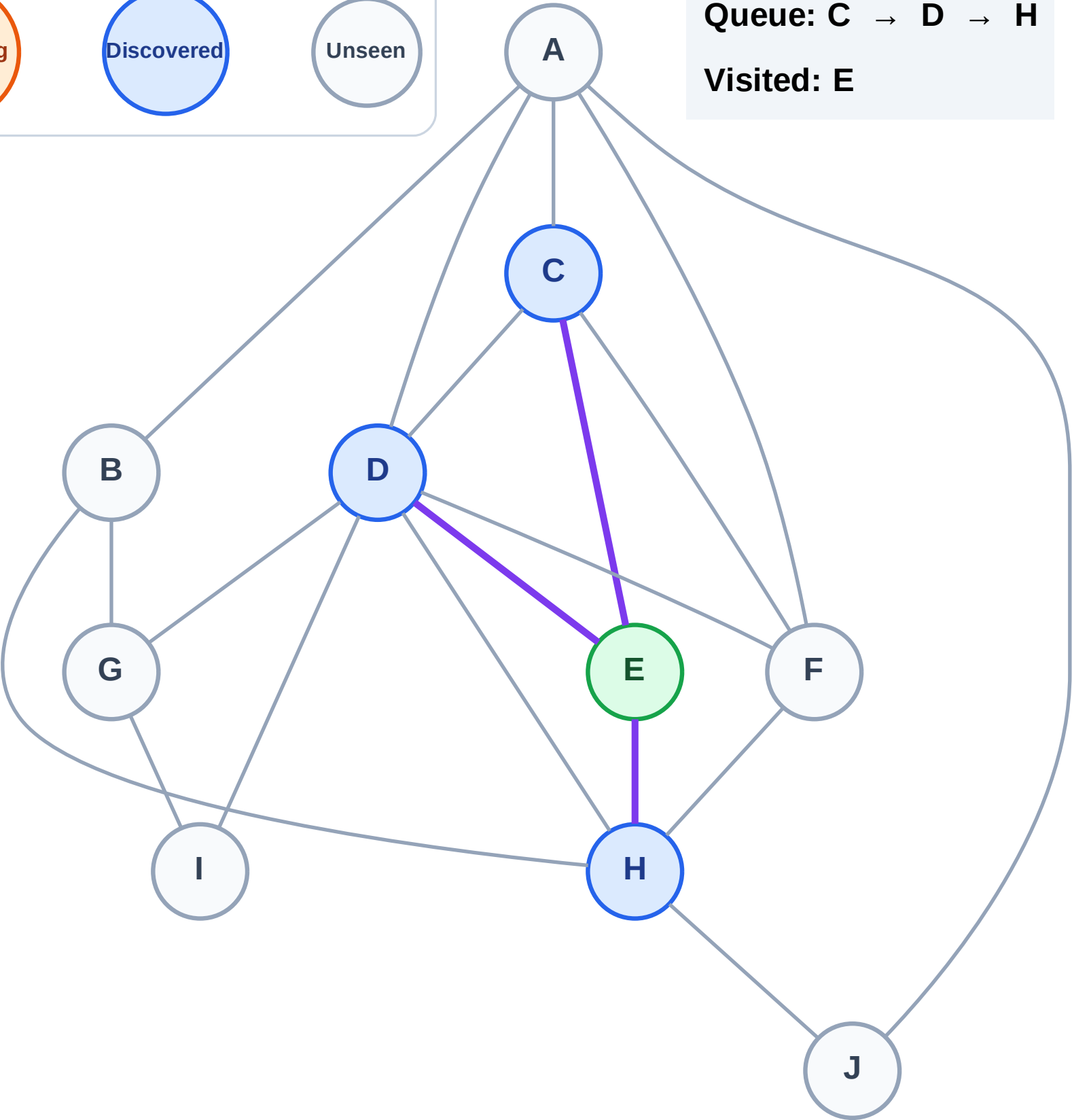
Processing

Discovered

Unseen

Queue: C → D → H

Visited: E



# BFS Traversal

Step 4: Process node C

Legend

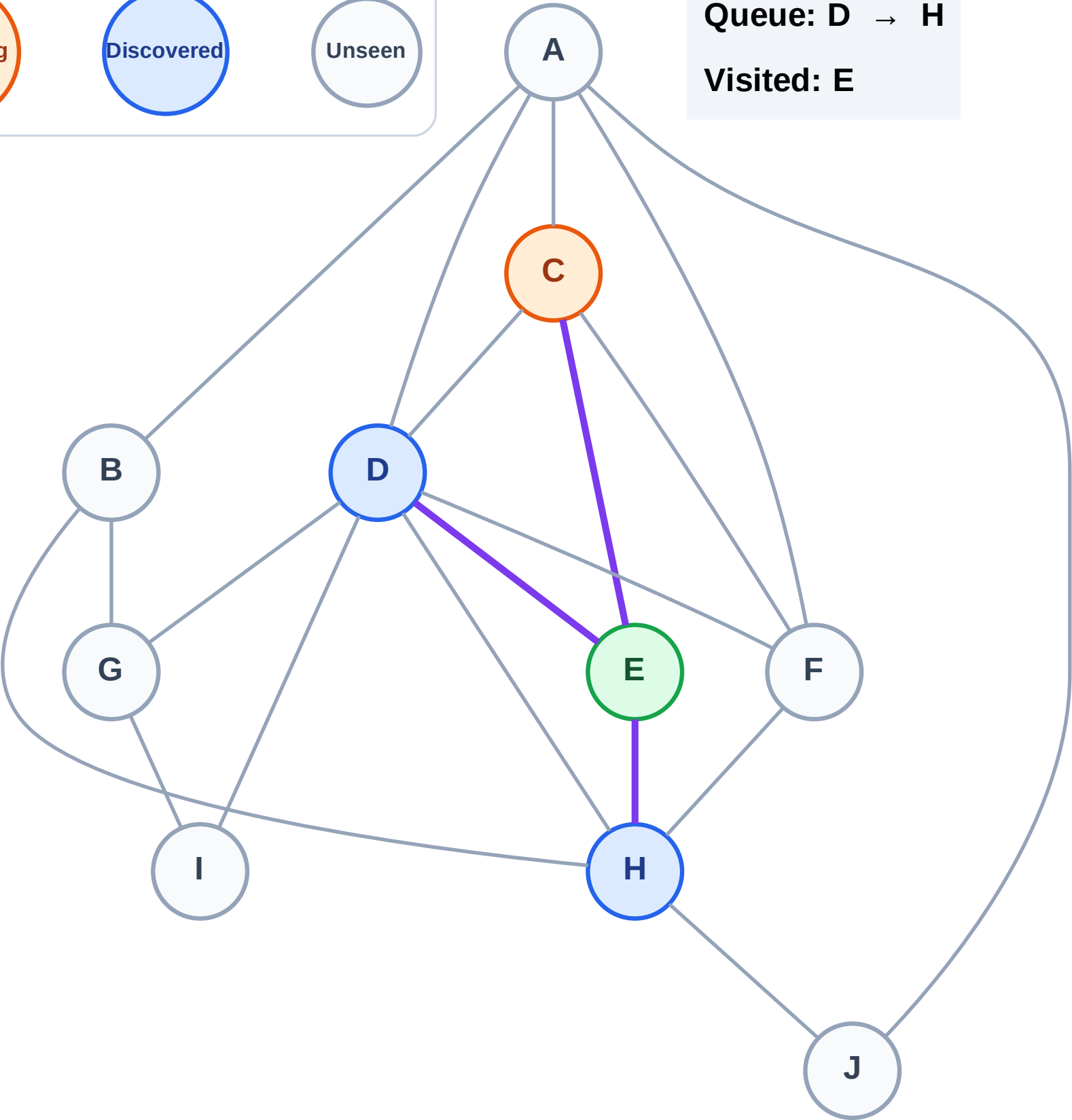
Complete

Processing

Discovered

Unseen

Queue: D → H  
Visited: E



# BFS Traversal

Step 5: Discover A, F from C

## Legend

Complete

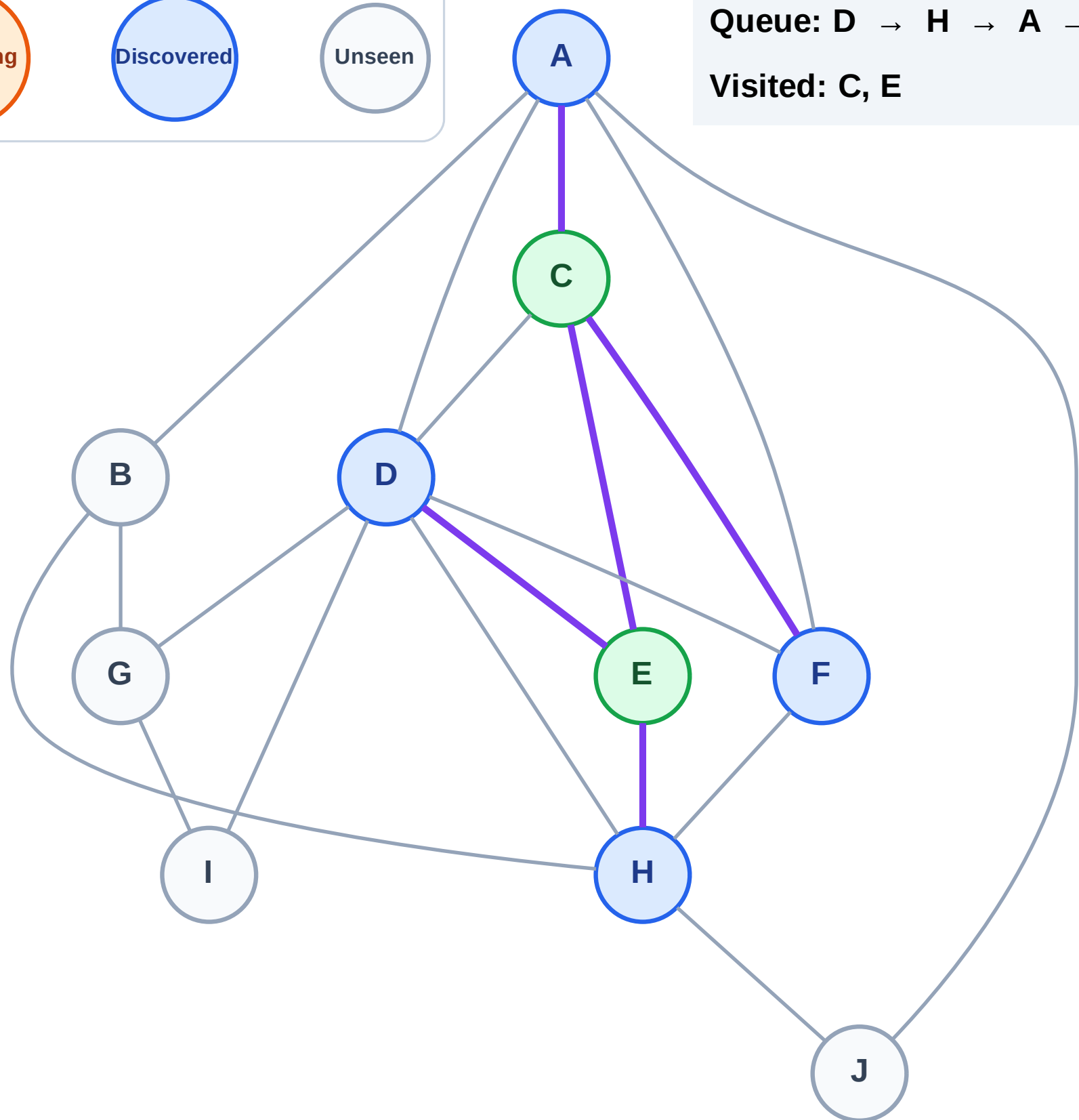
Processing

Discovered

Unseen

Queue: D → H → A → F

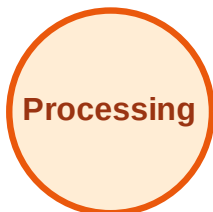
Visited: C, E



# BFS Traversal

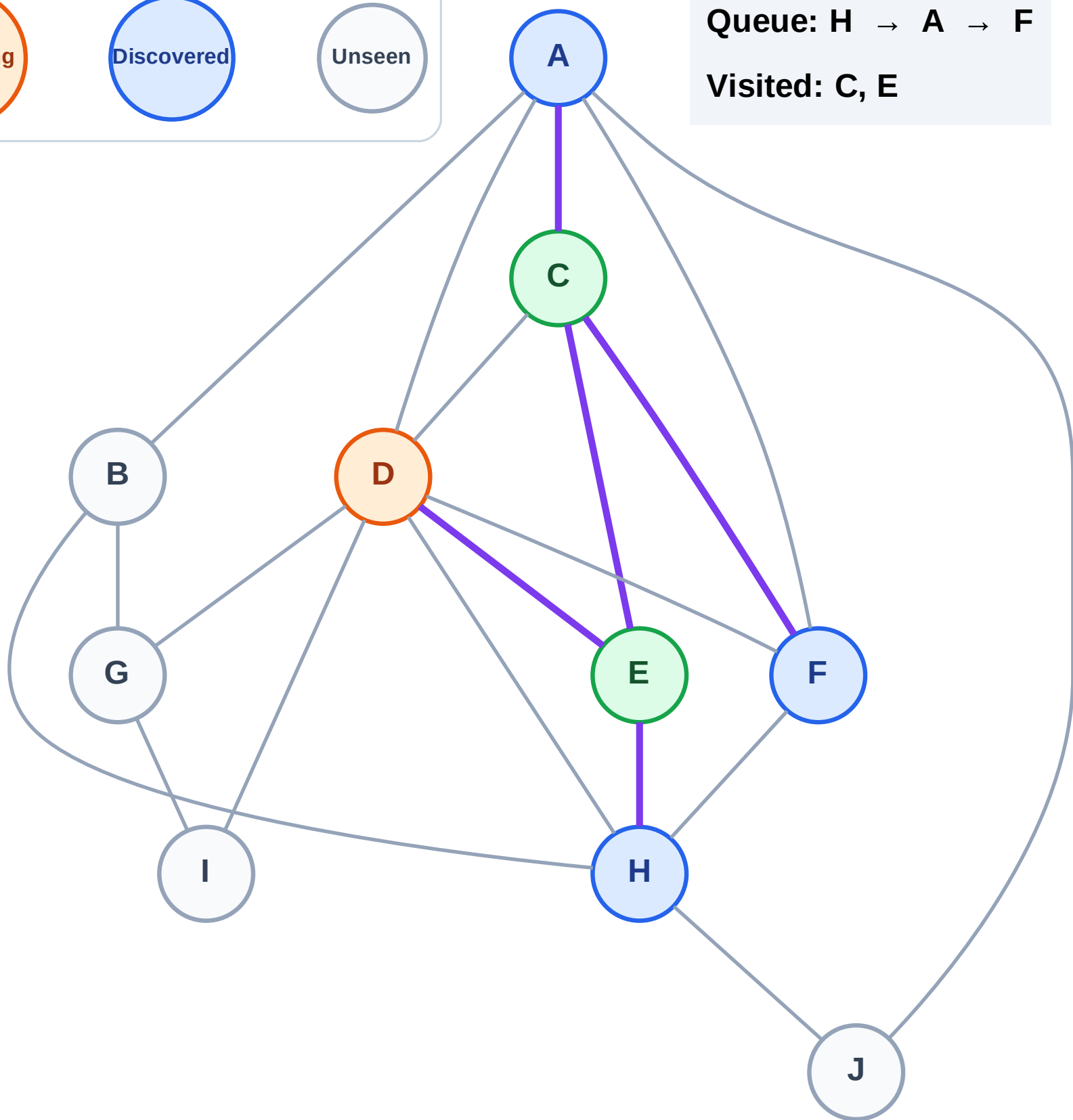
Step 6: Process node D

## Legend



Queue: H → A → F

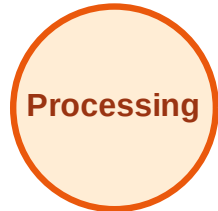
Visited: C, E



# BFS Traversal

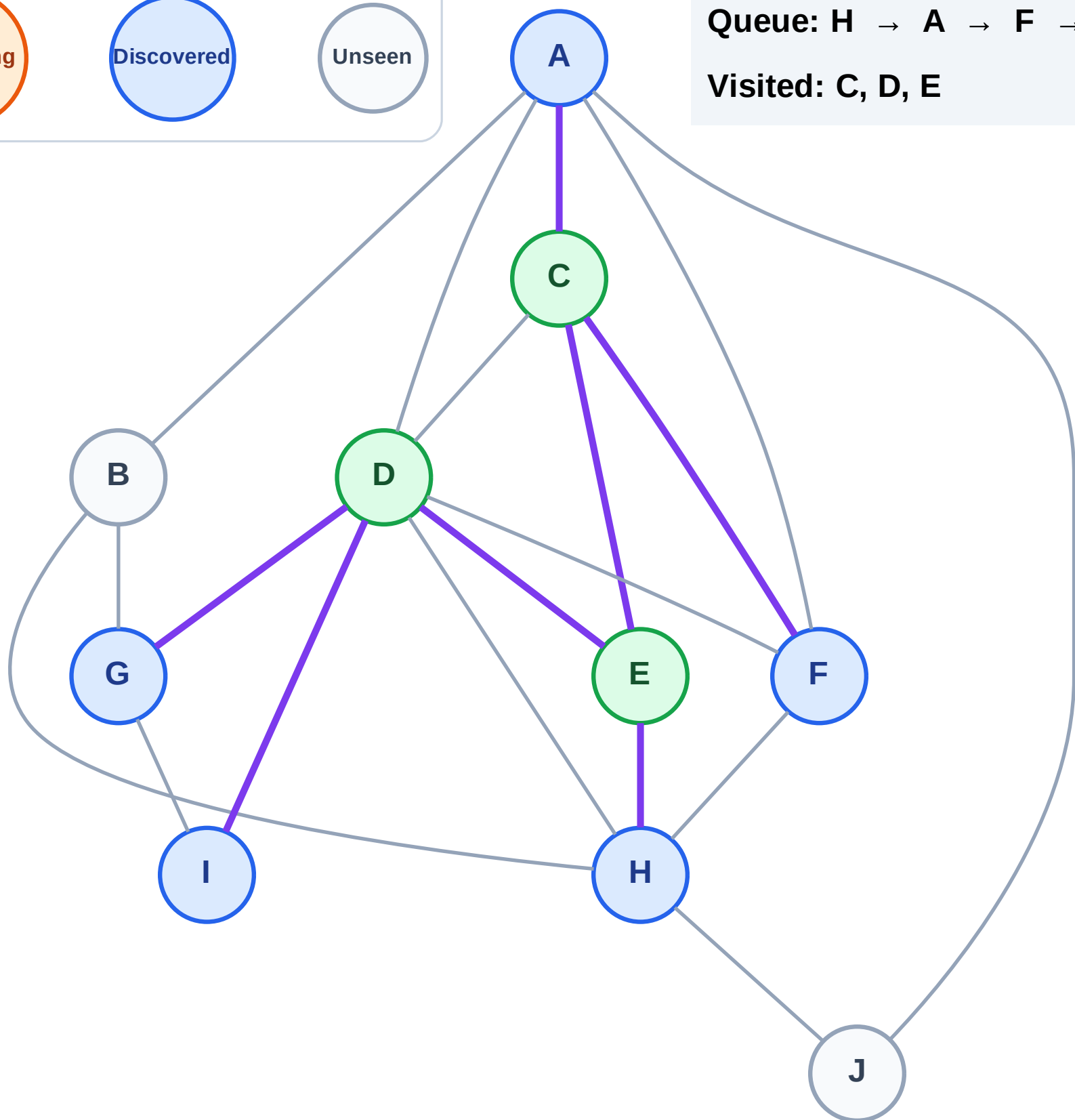
Step 7: Discover G, I from D

## Legend



Queue: H → A → F → G → I

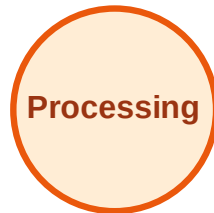
Visited: C, D, E



# BFS Traversal

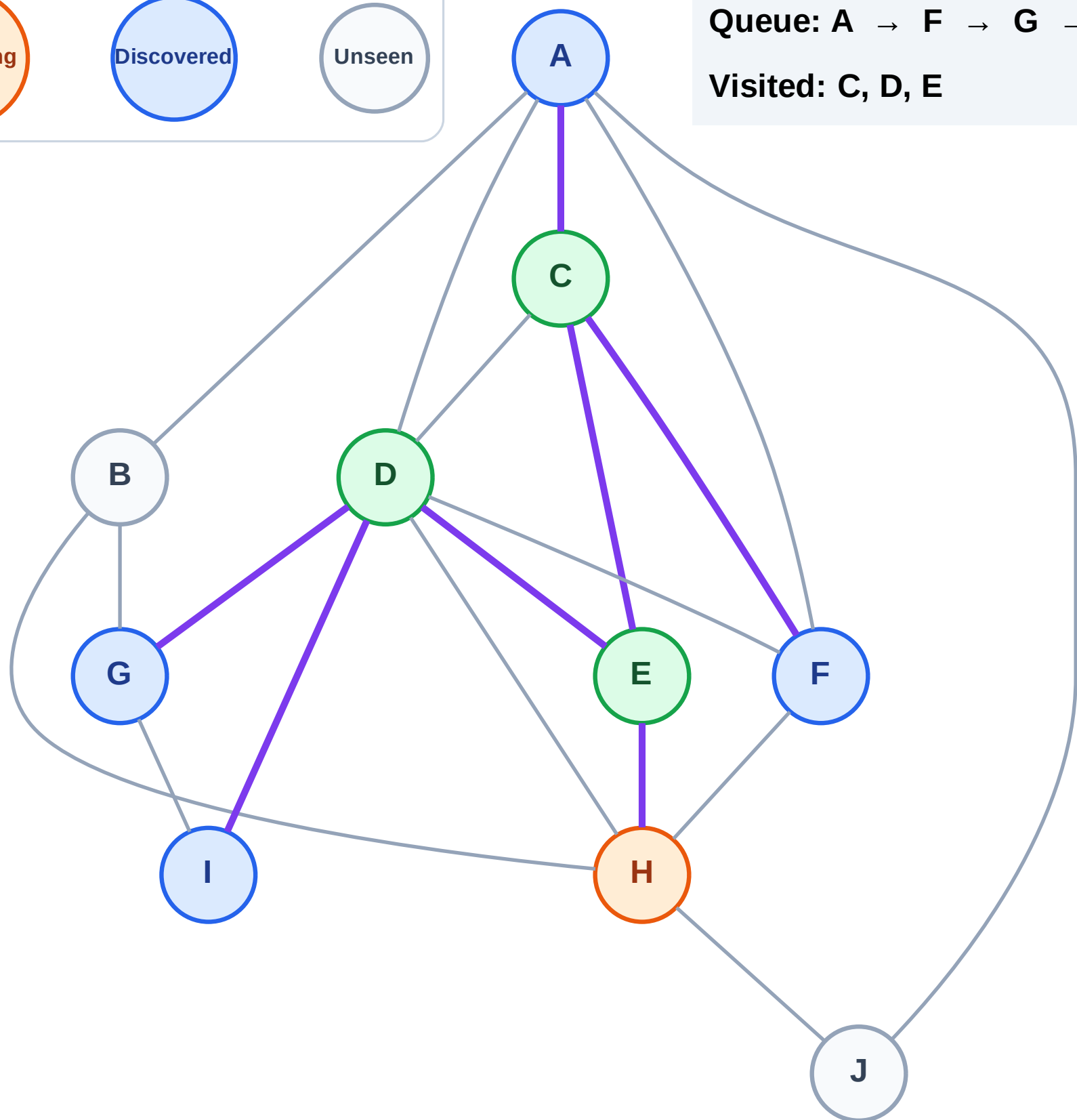
Step 8: Process node H

## Legend



Queue: A → F → G → I

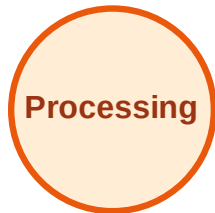
Visited: C, D, E



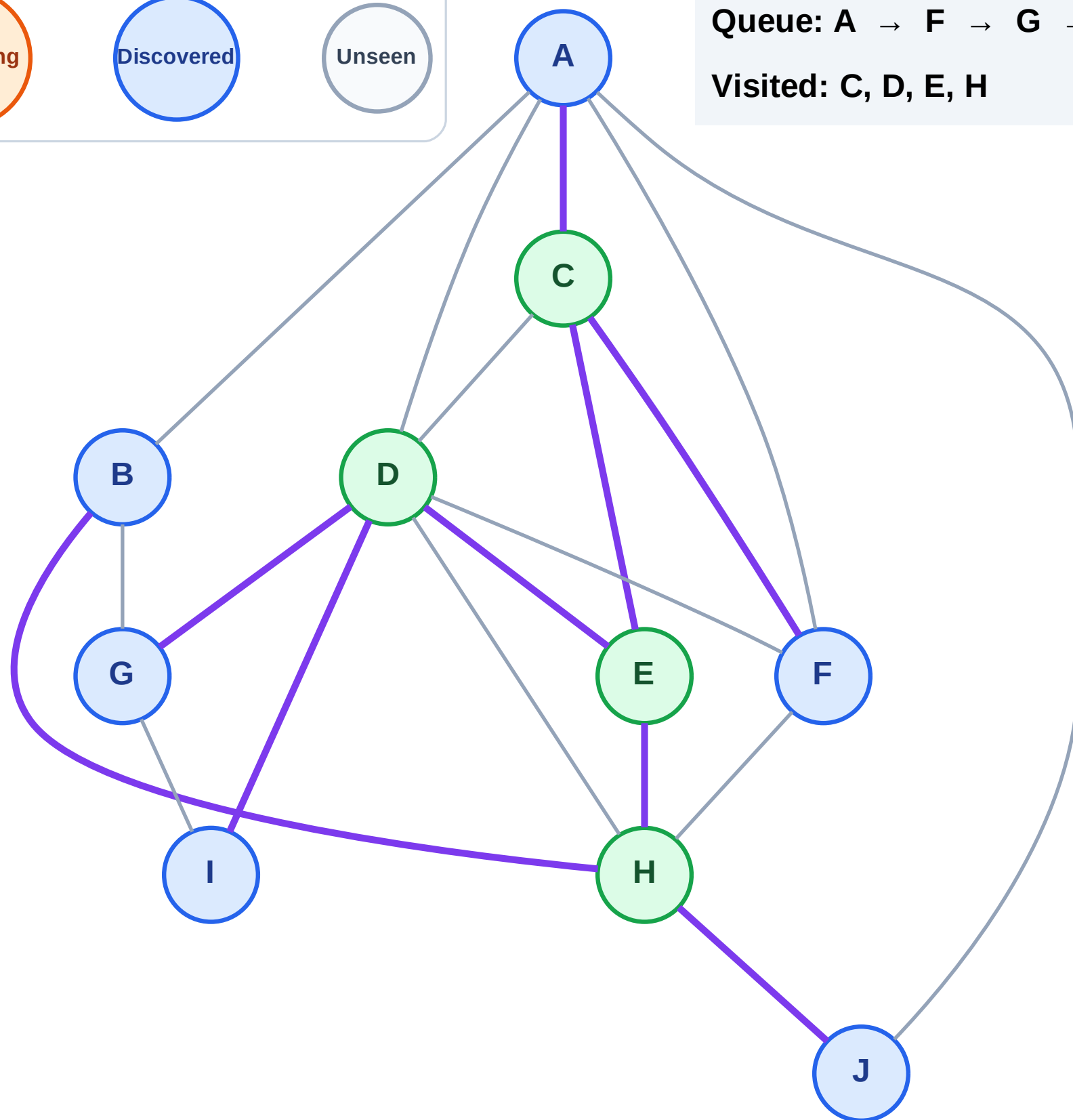
# BFS Traversal

Step 9: Discover B, J from H

## Legend



Queue: A → F → G → I → B → J  
Visited: C, D, E, H



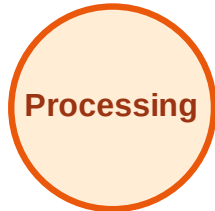
# BFS Traversal

Step 10: Process node A

## Legend



Complete



Processing



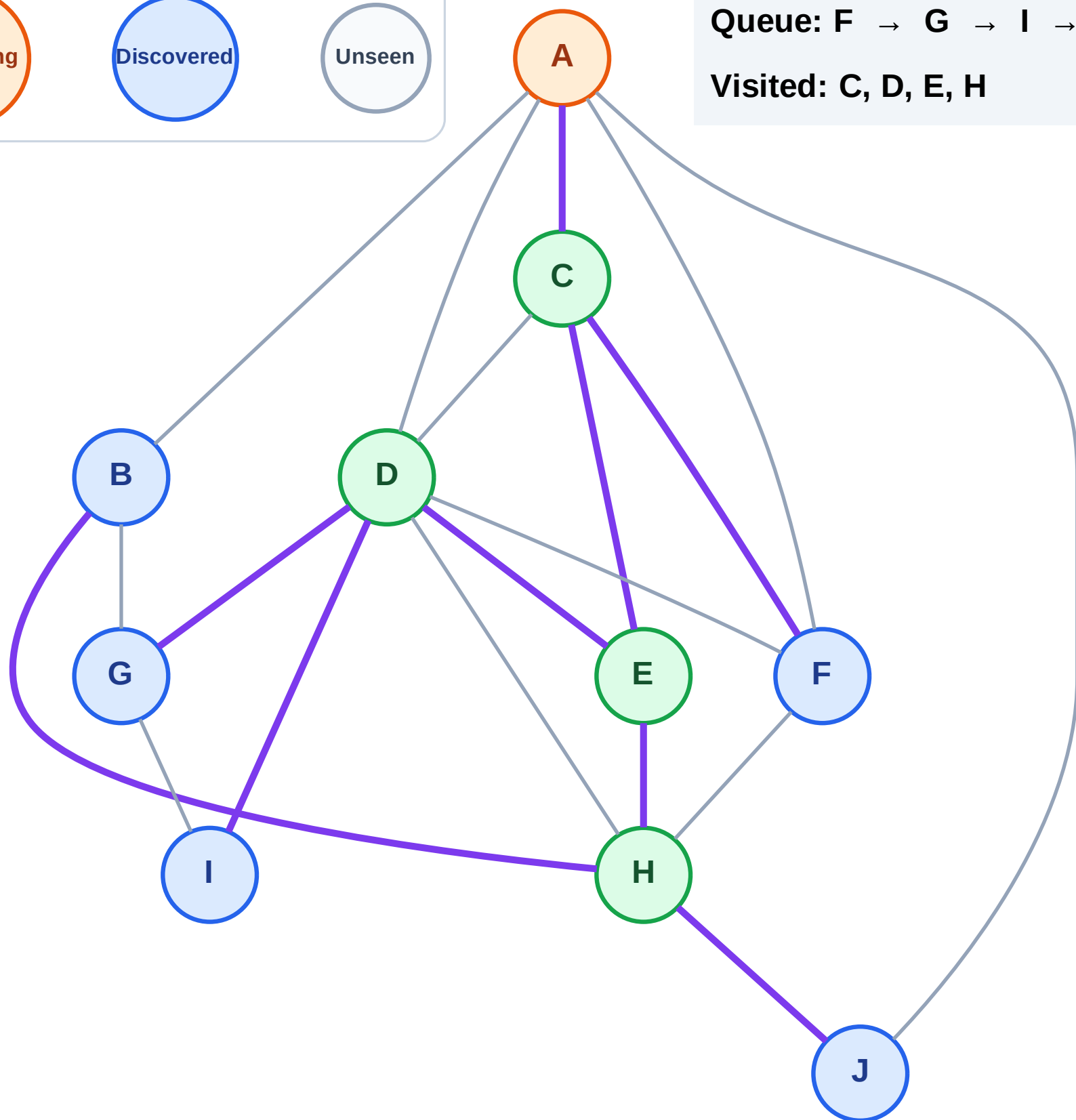
Discovered



Unseen

Queue: F → G → I → B → J

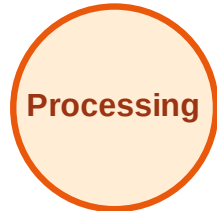
Visited: C, D, E, H



# BFS Traversal

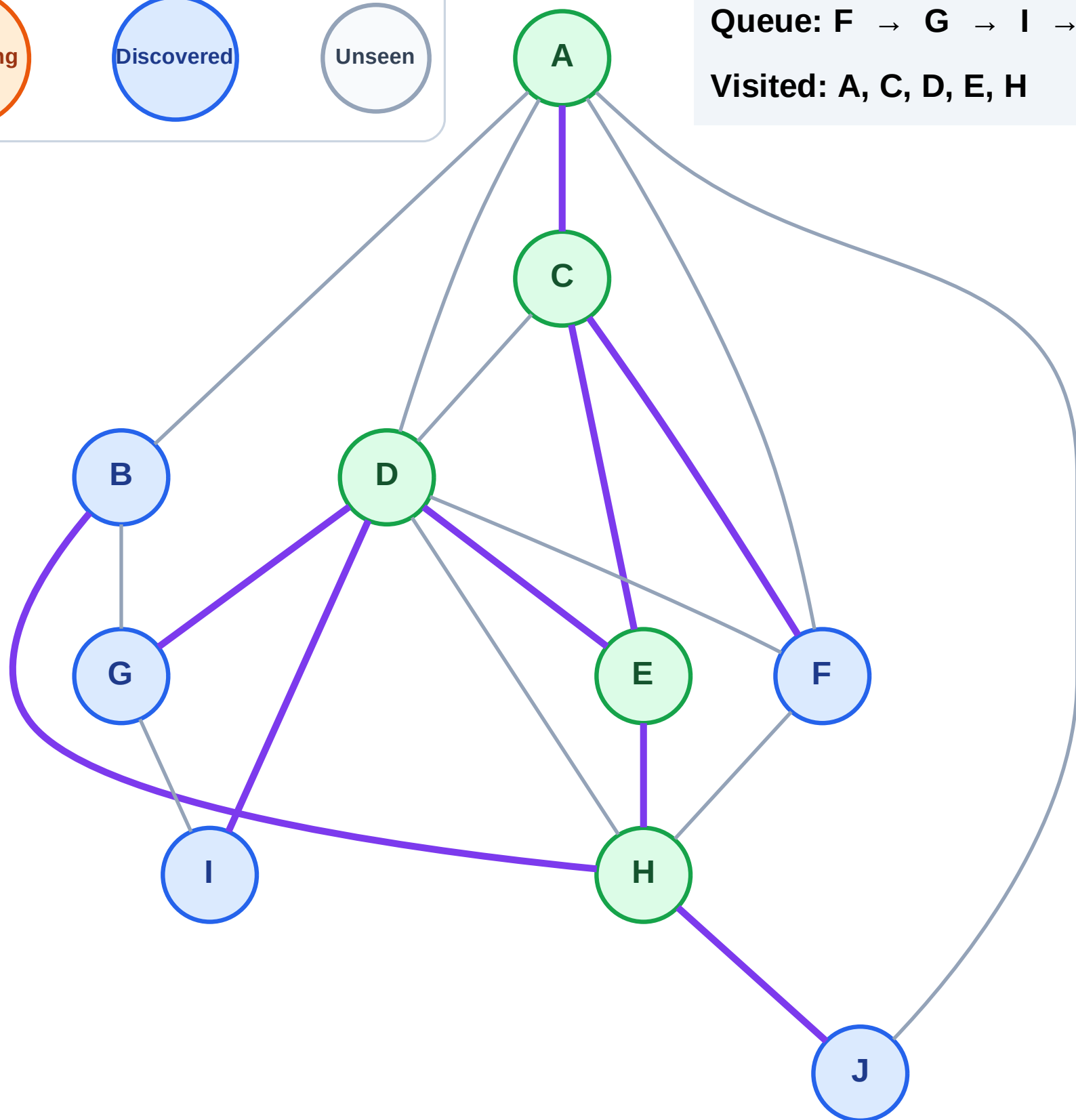
Step 11: No new neighbors from A

## Legend



Queue: F → G → I → B → J

Visited: A, C, D, E, H



# BFS Traversal

Step 12: Process node F

Legend

Complete

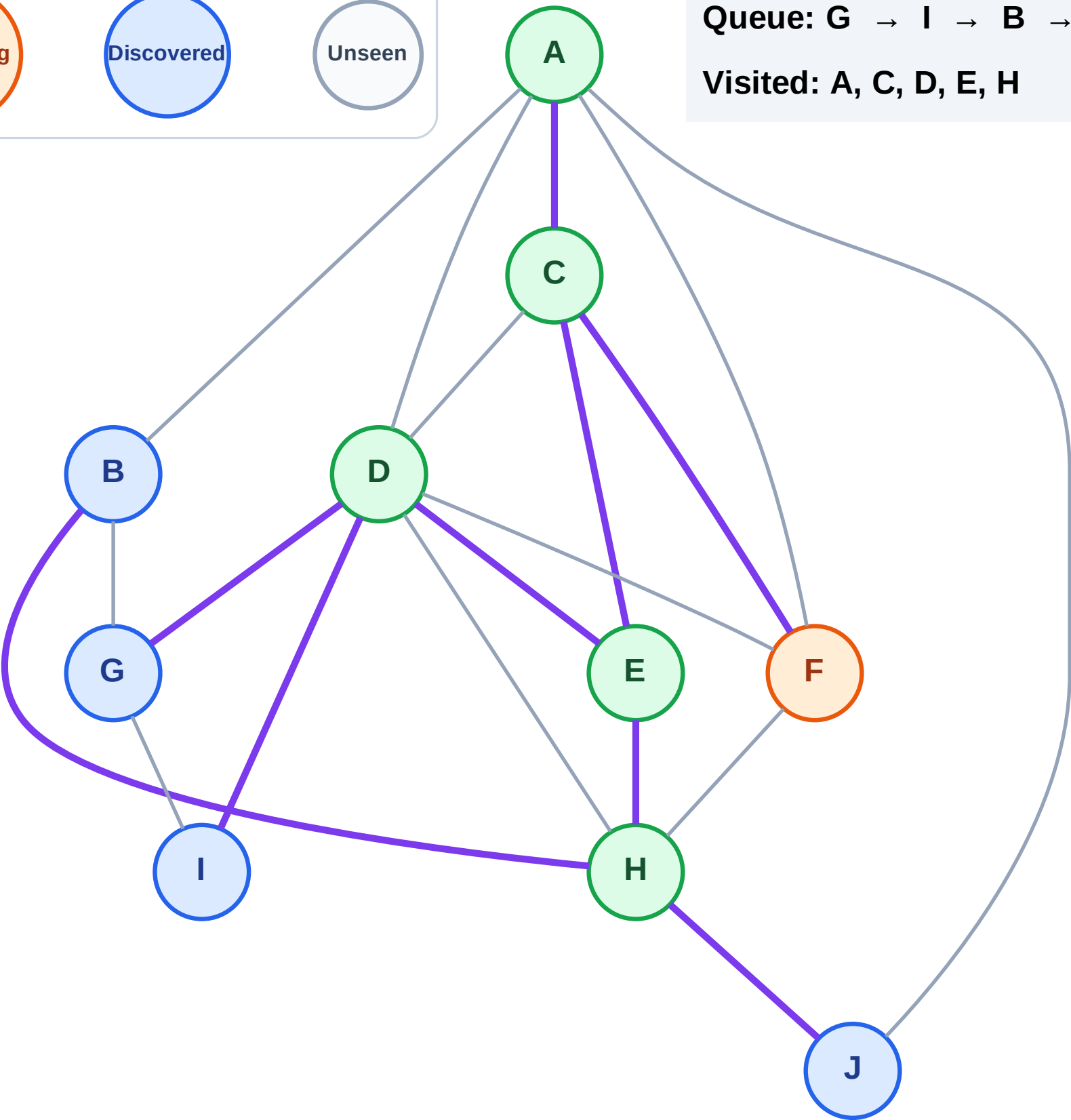
Processing

Discovered

Unseen

Queue: G → I → B → J

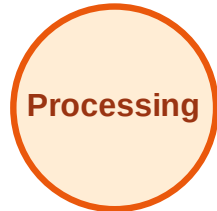
Visited: A, C, D, E, H



# BFS Traversal

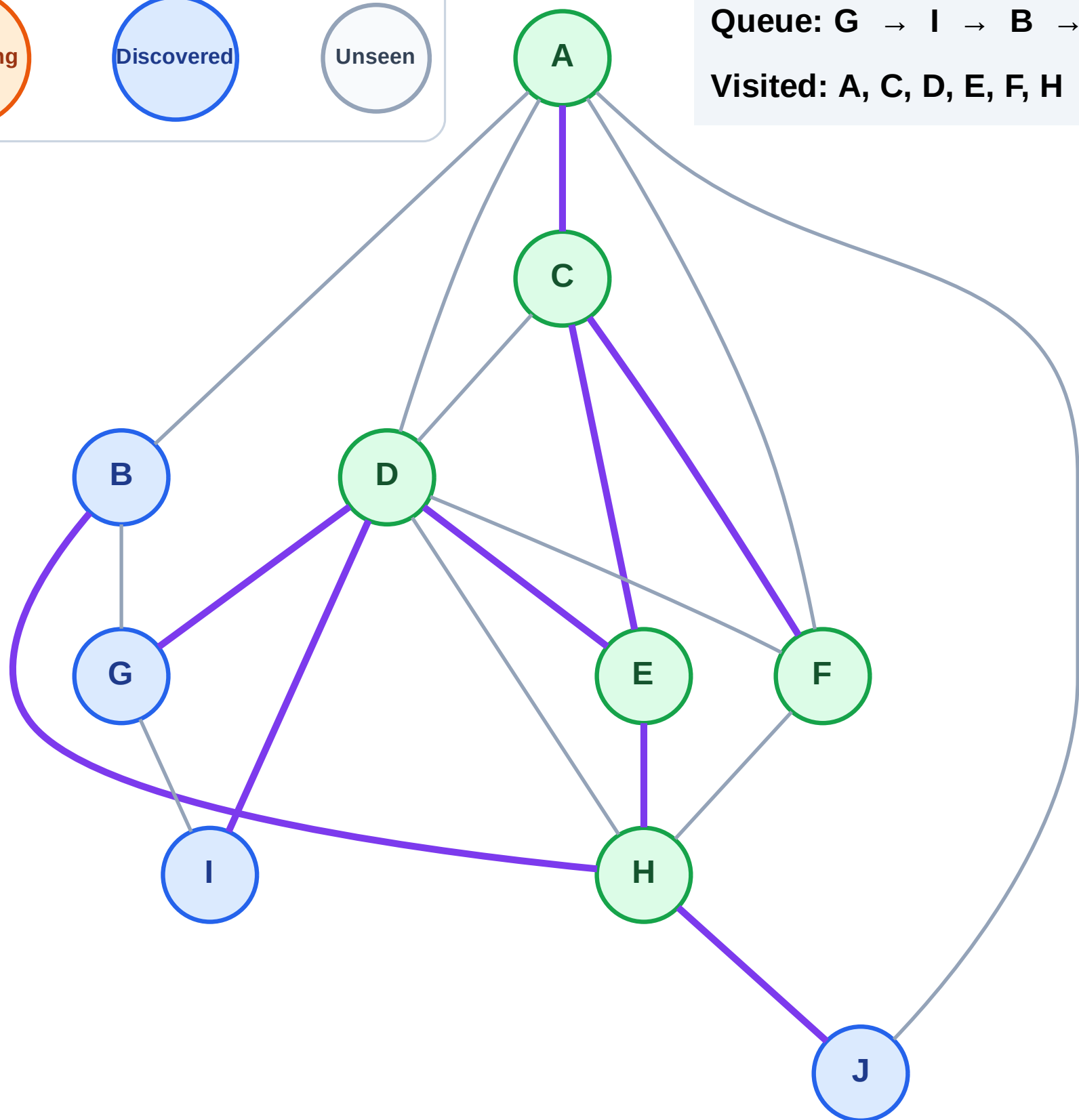
Step 13: No new neighbors from F

## Legend



Queue: G → I → B → J

Visited: A, C, D, E, F, H



# BFS Traversal

Step 14: Process node G

Legend

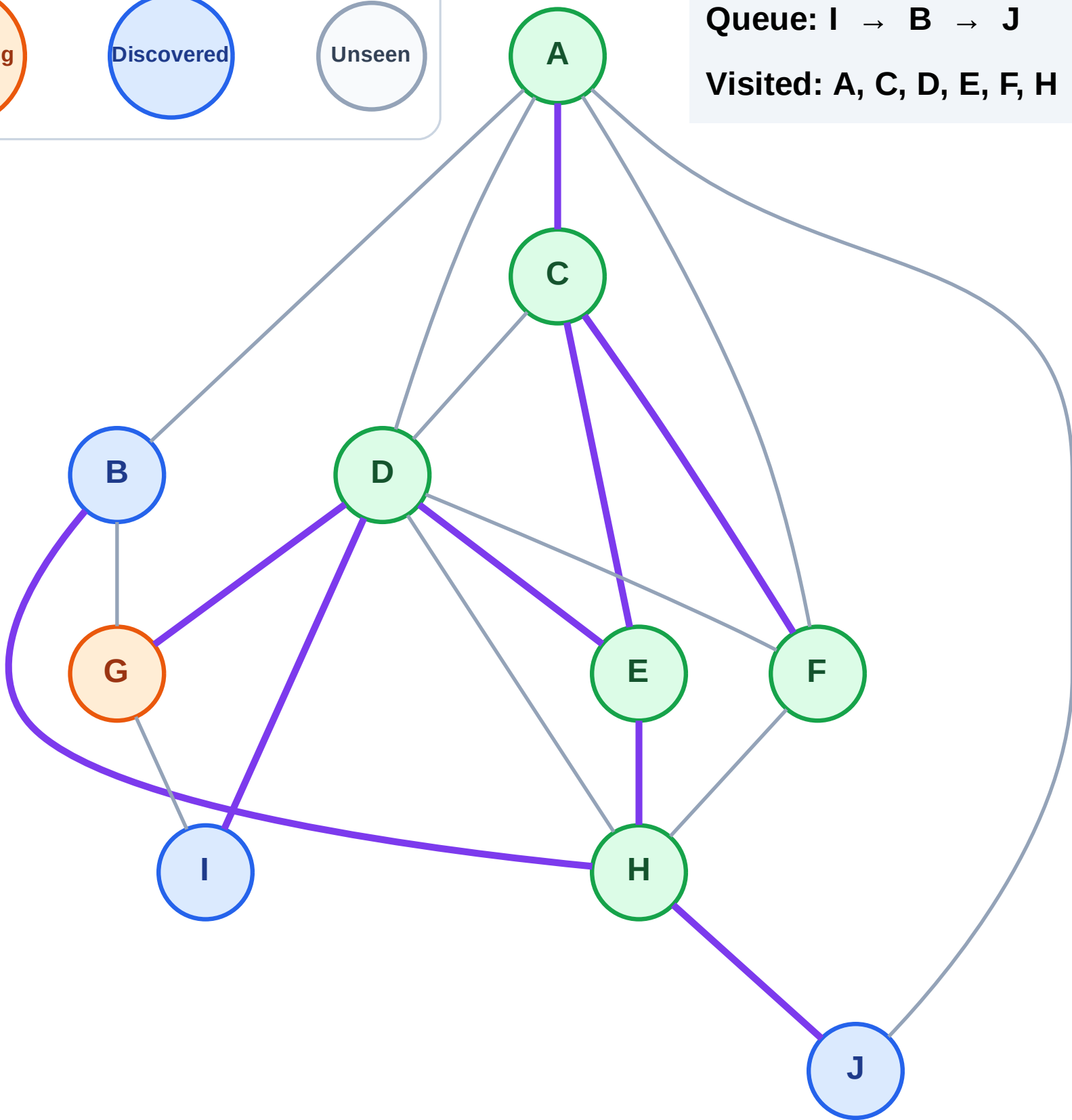
Complete

Processing

Discovered

Unseen

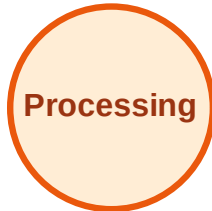
Queue: I → B → J  
Visited: A, C, D, E, F, H



# BFS Traversal

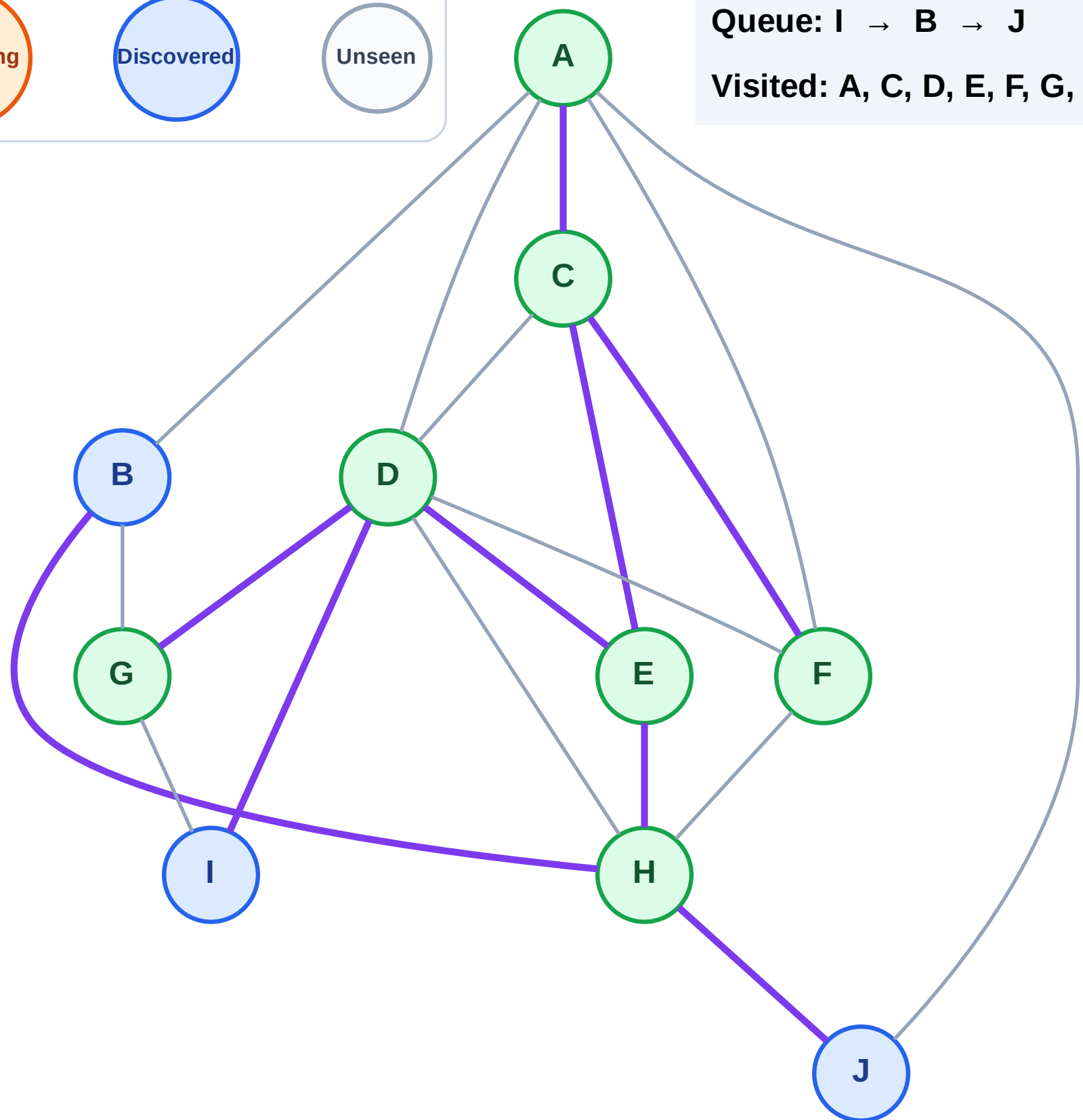
Step 15: No new neighbors from G

## Legend



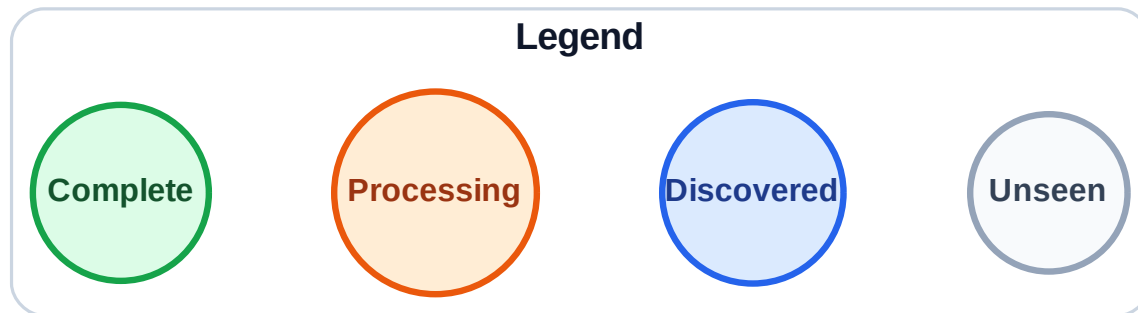
Queue: I → B → J

Visited: A, C, D, E, F, G, H



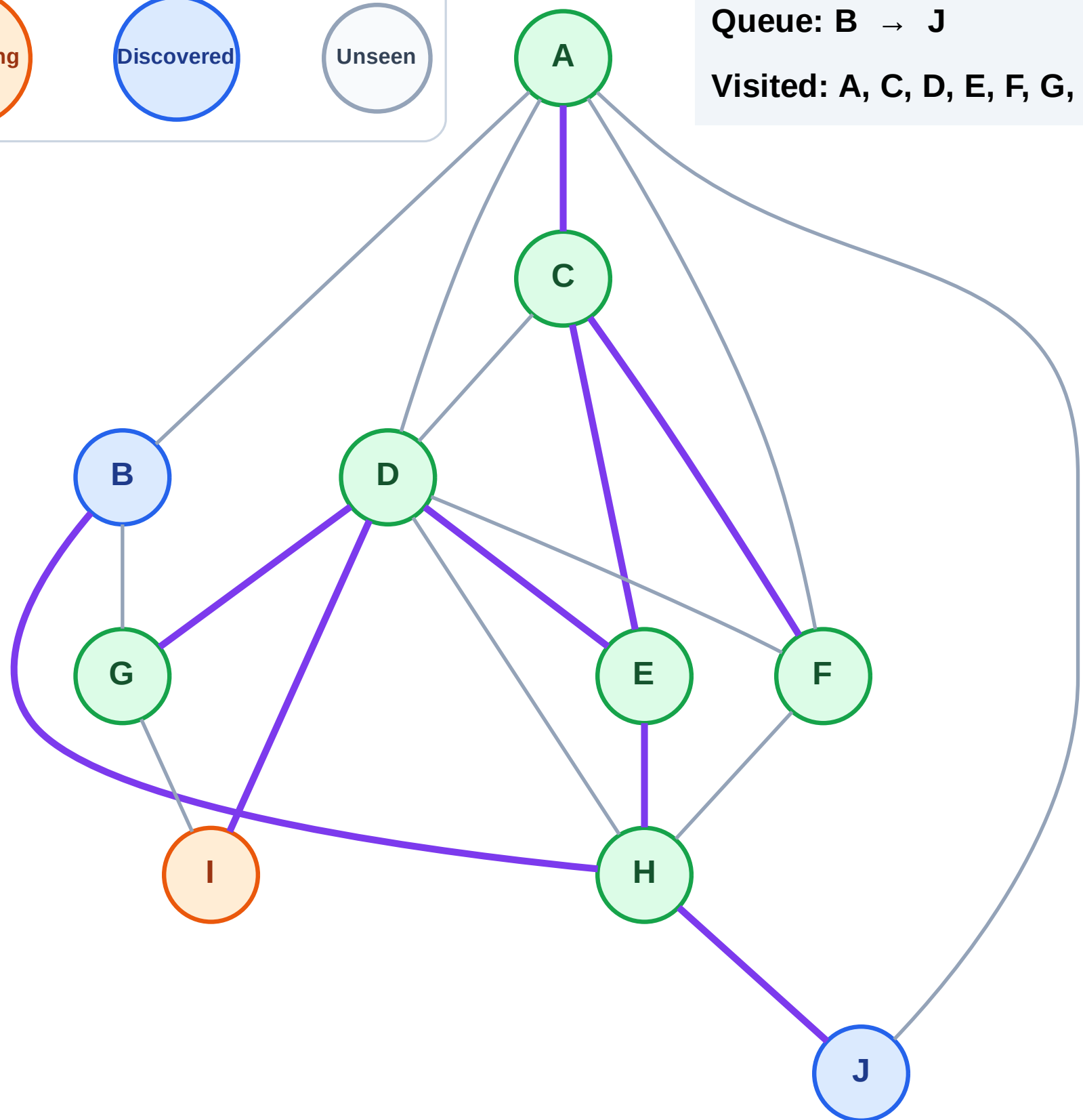
### Step 16: Process node I

### Step 16: Process node I



**Queue: B → J**

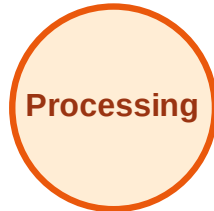
**Visited: A, C, D, E, F, G, H**



# BFS Traversal

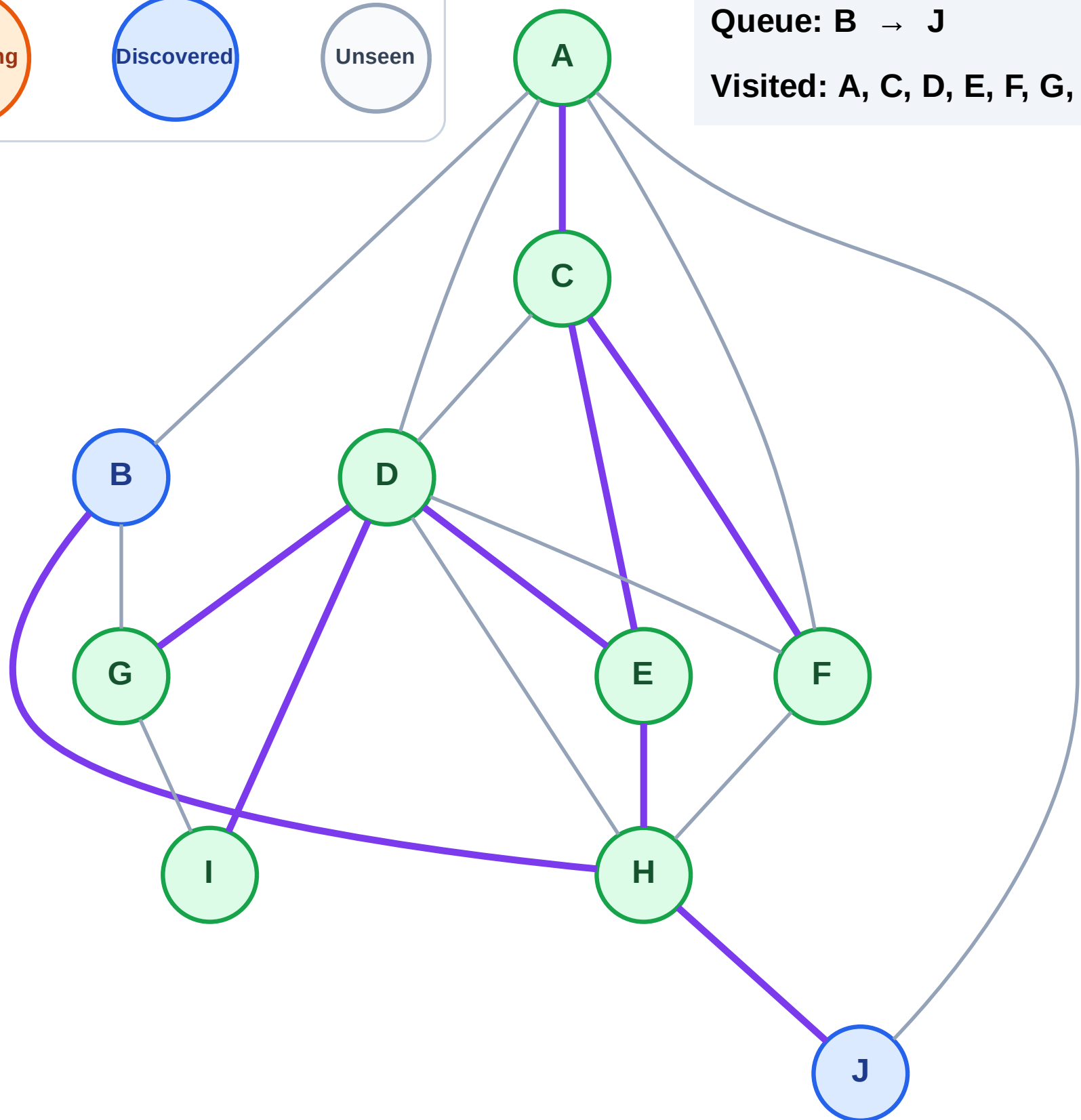
Step 17: No new neighbors from I

## Legend



Queue: B → J

Visited: A, C, D, E, F, G, H, I



# BFS Traversal

Step 18: Process node B

Legend

Complete

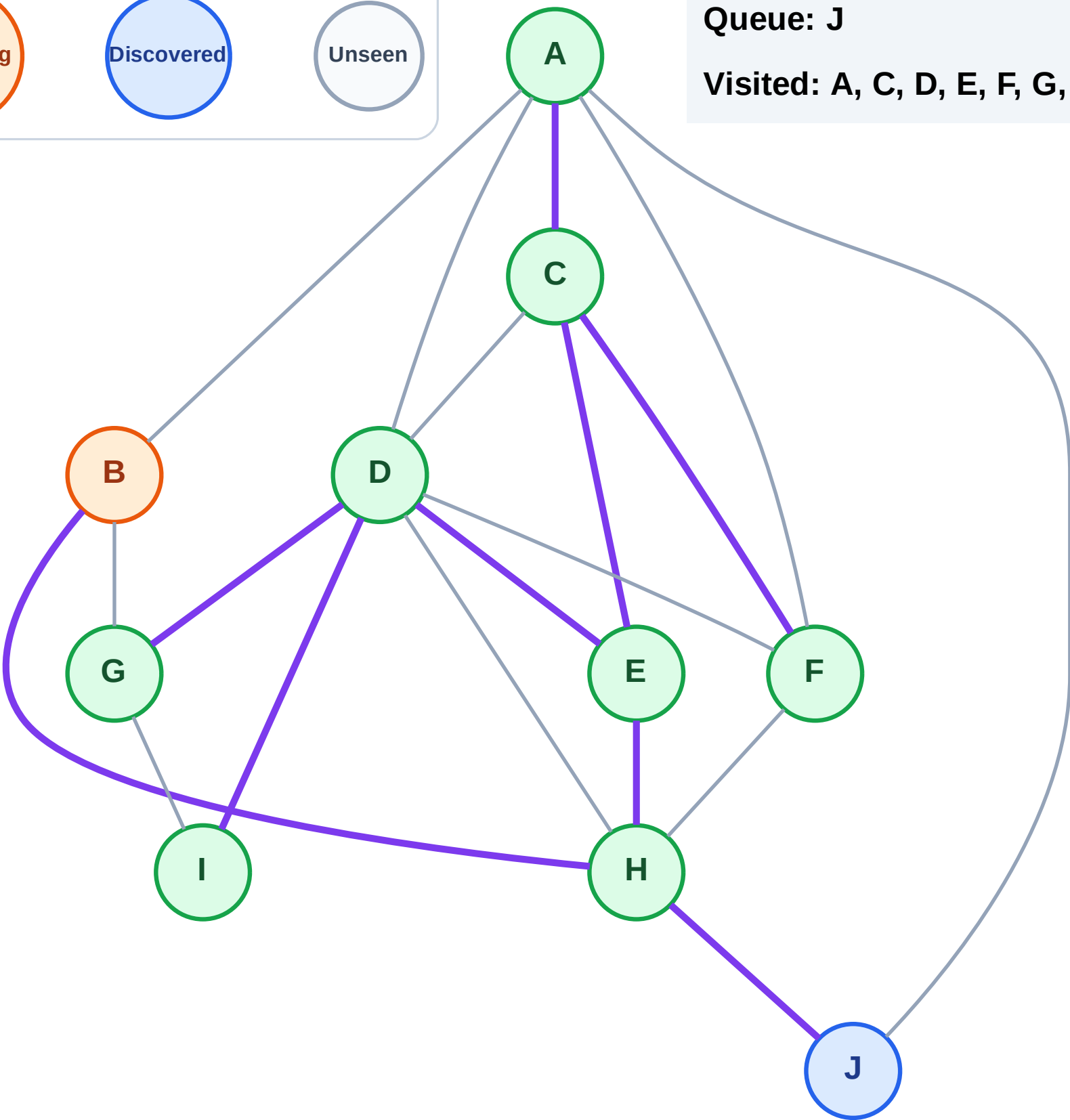
Processing

Discovered

Unseen

Queue: J

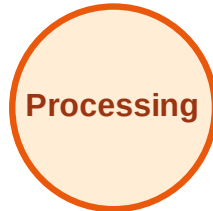
Visited: A, C, D, E, F, G, H, I



# BFS Traversal

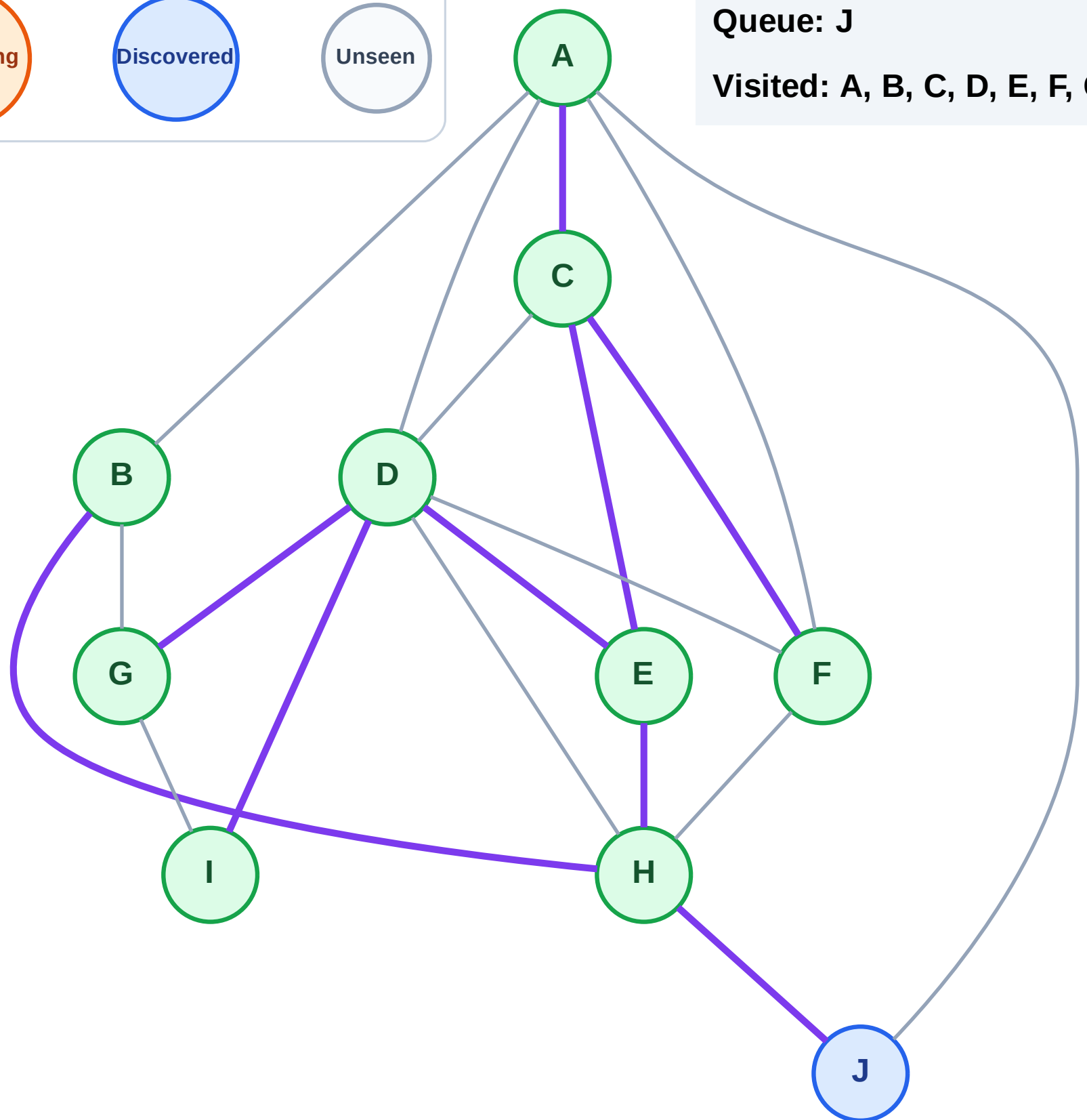
Step 19: No new neighbors from B

## Legend



Queue: J

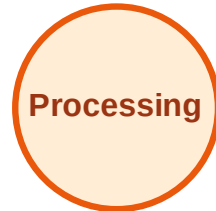
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

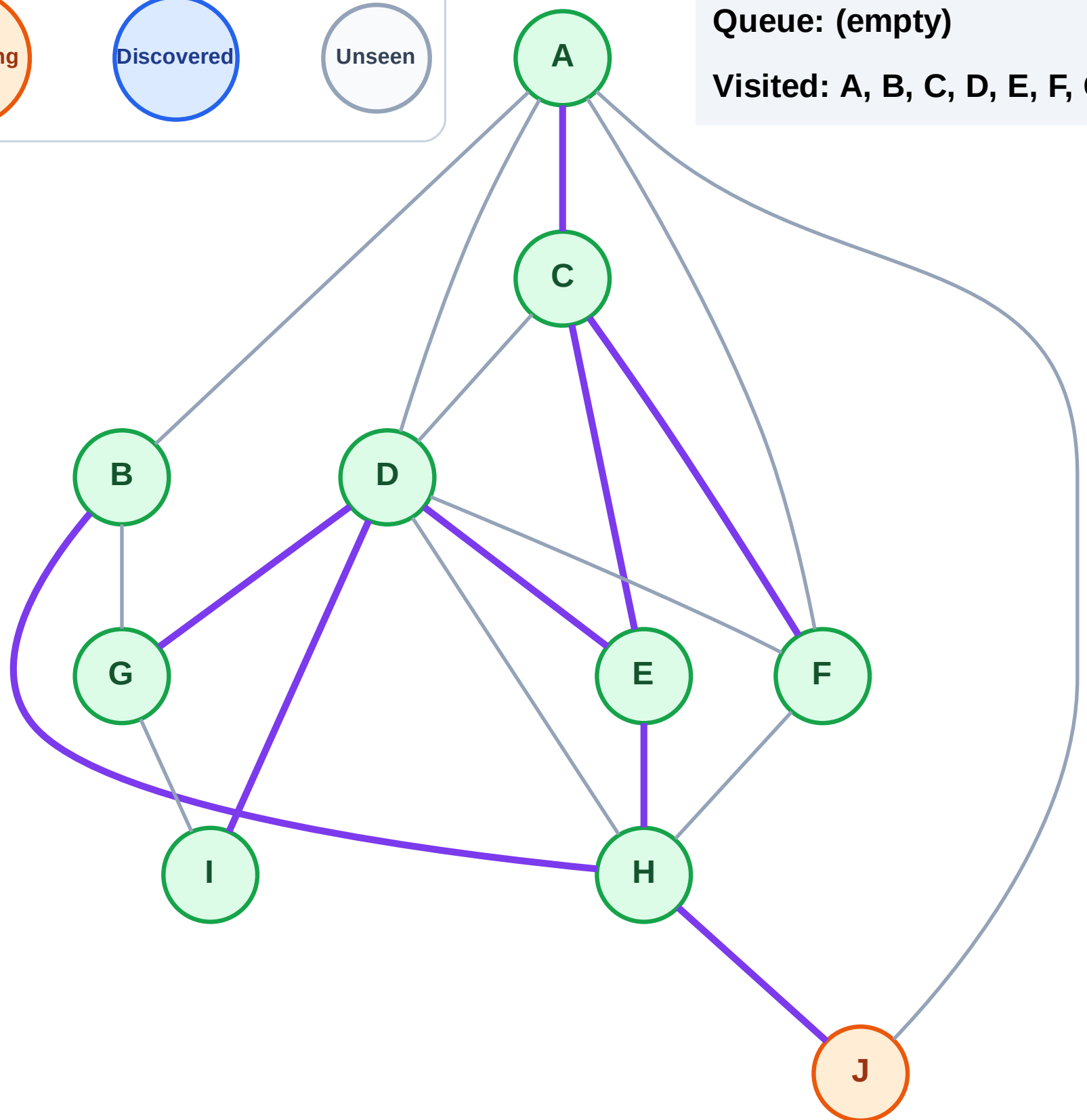
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

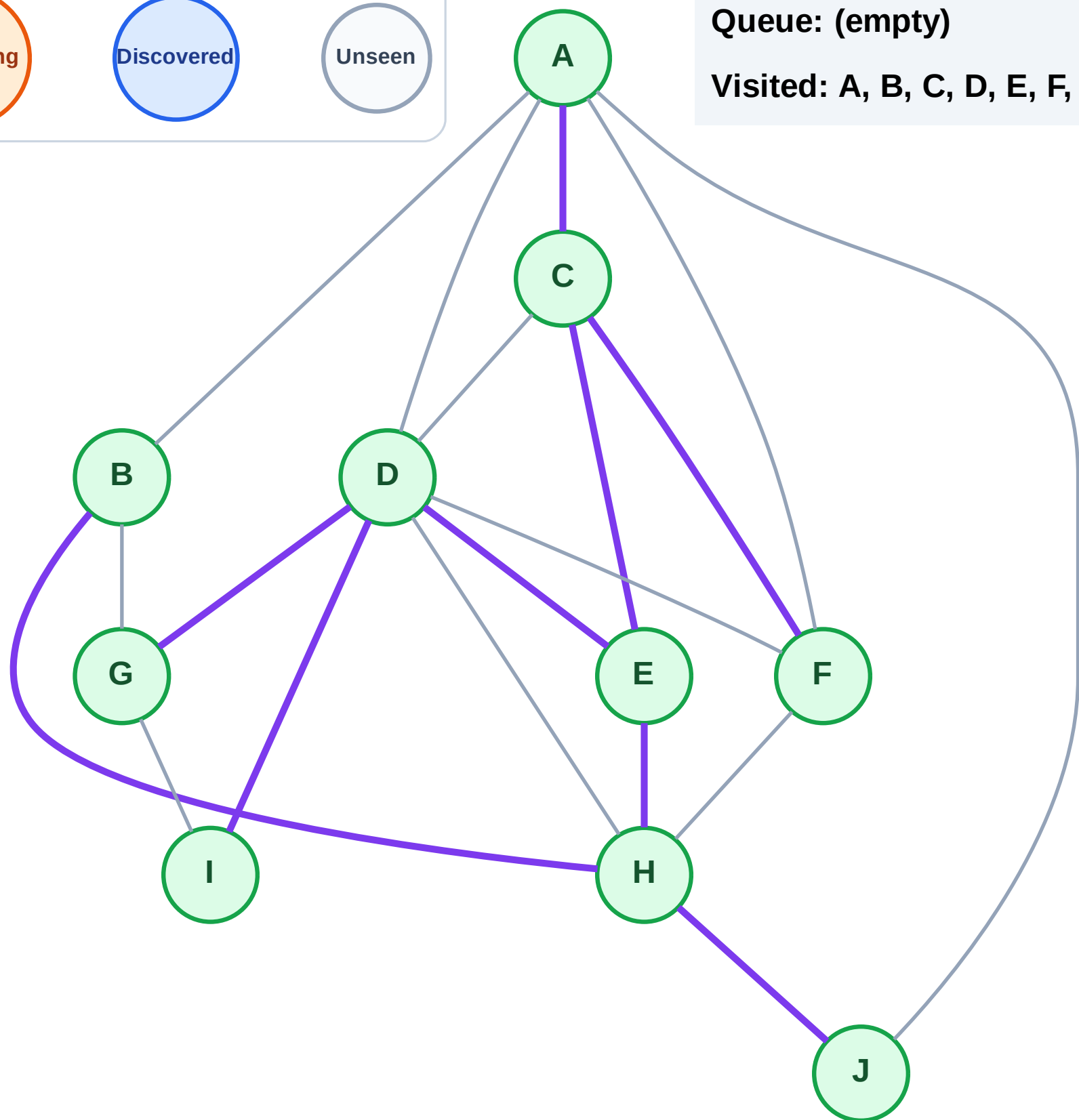
Step 21: No new neighbors from J

## Legend



Queue: (empty)

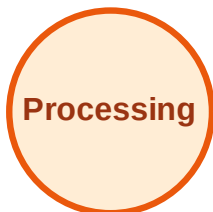
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

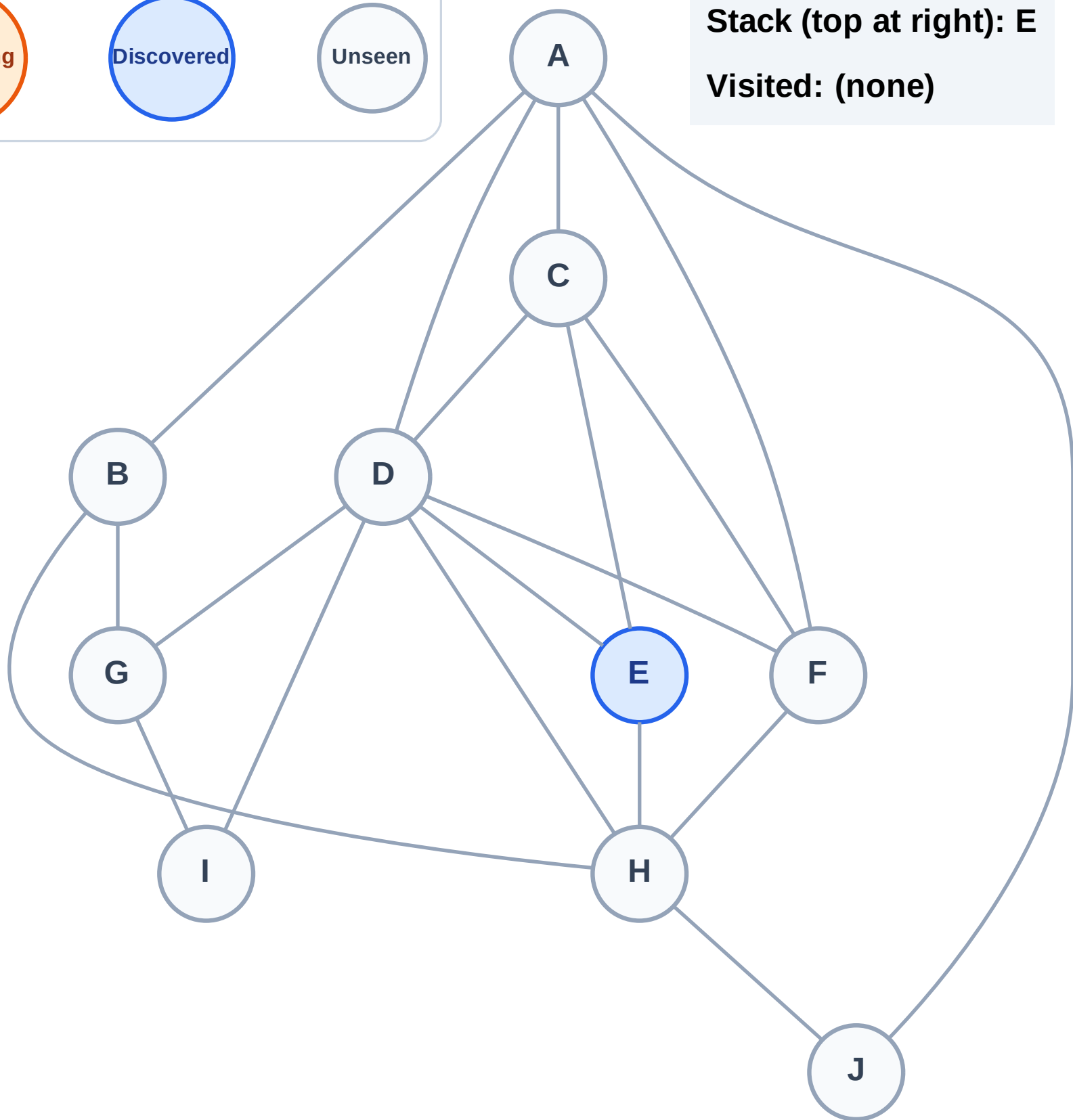
Step 1: Start at E

## Legend



Stack (top at right): E

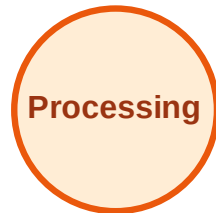
Visited: (none)



# DFS Traversal

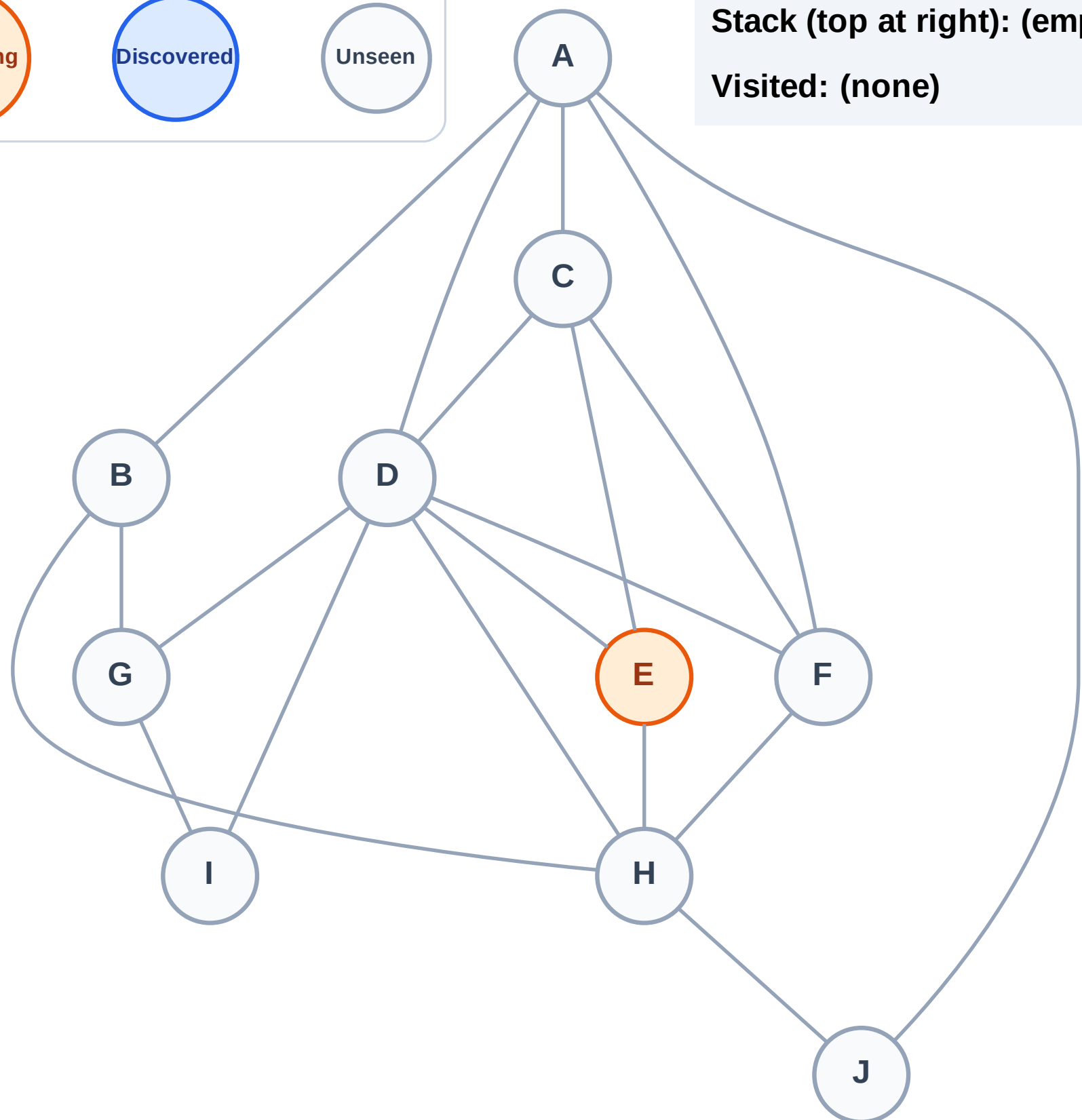
Step 2: Process node E

## Legend



Stack (top at right): (empty)

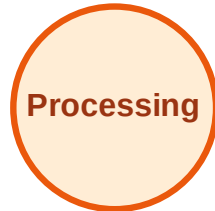
Visited: (none)



# DFS Traversal

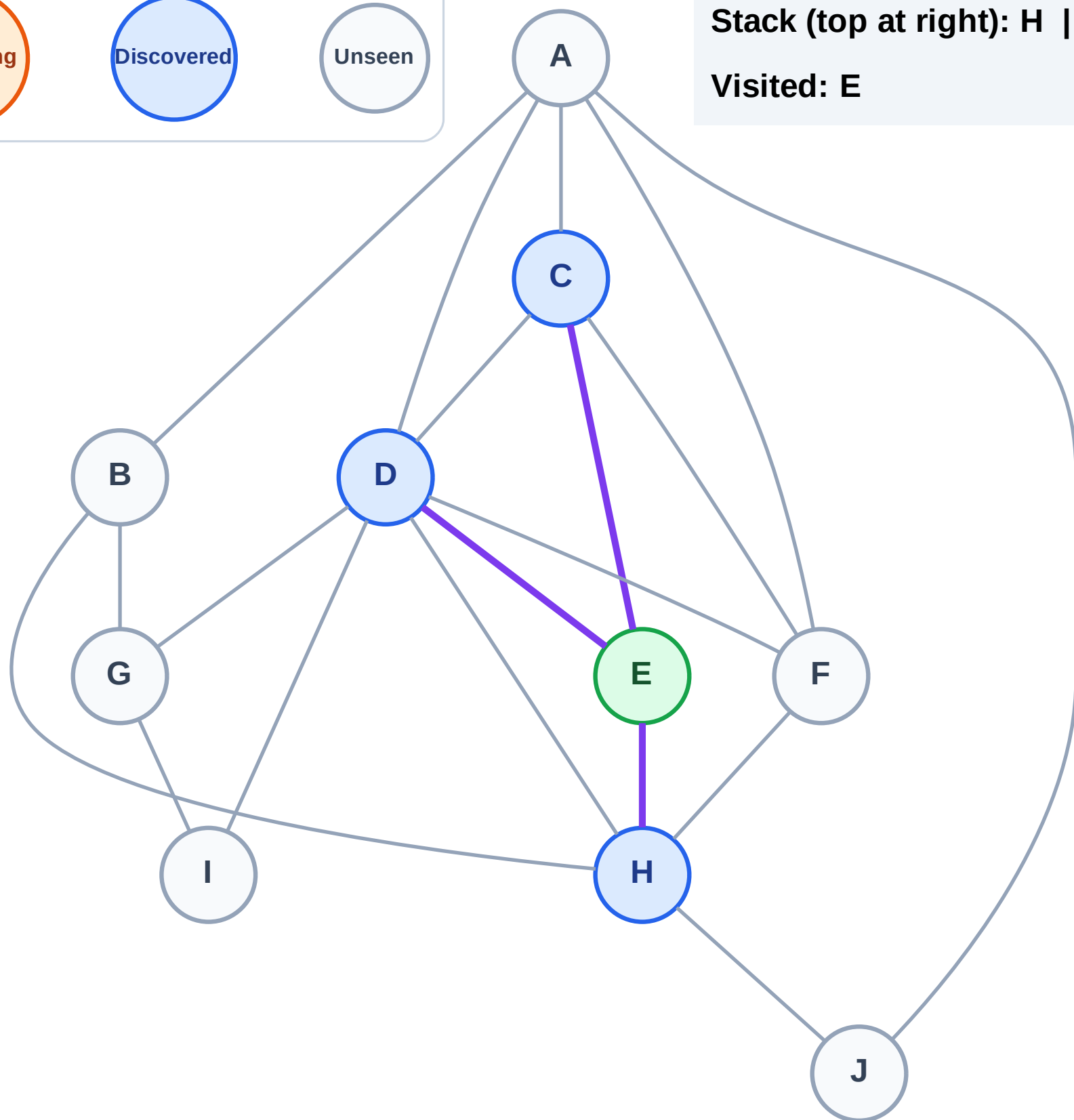
Step 3: Discover H, D, C from E

## Legend



Stack (top at right): H | D | C

Visited: E



# DFS Traversal

Step 4: Process node C

Legend

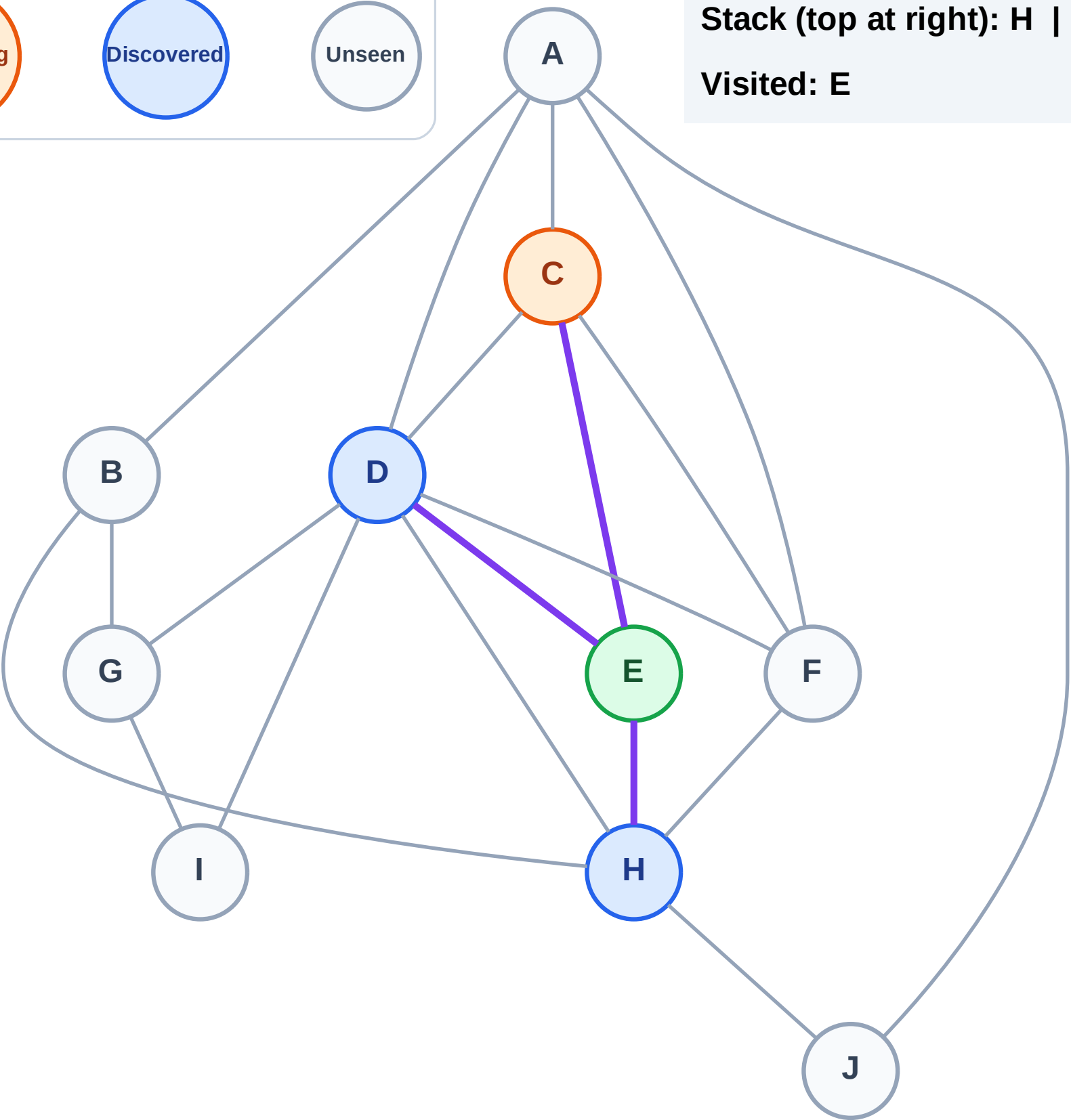
Complete

Processing

Discovered

Unseen

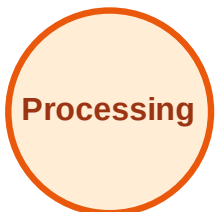
Stack (top at right): H | D  
Visited: E



# DFS Traversal

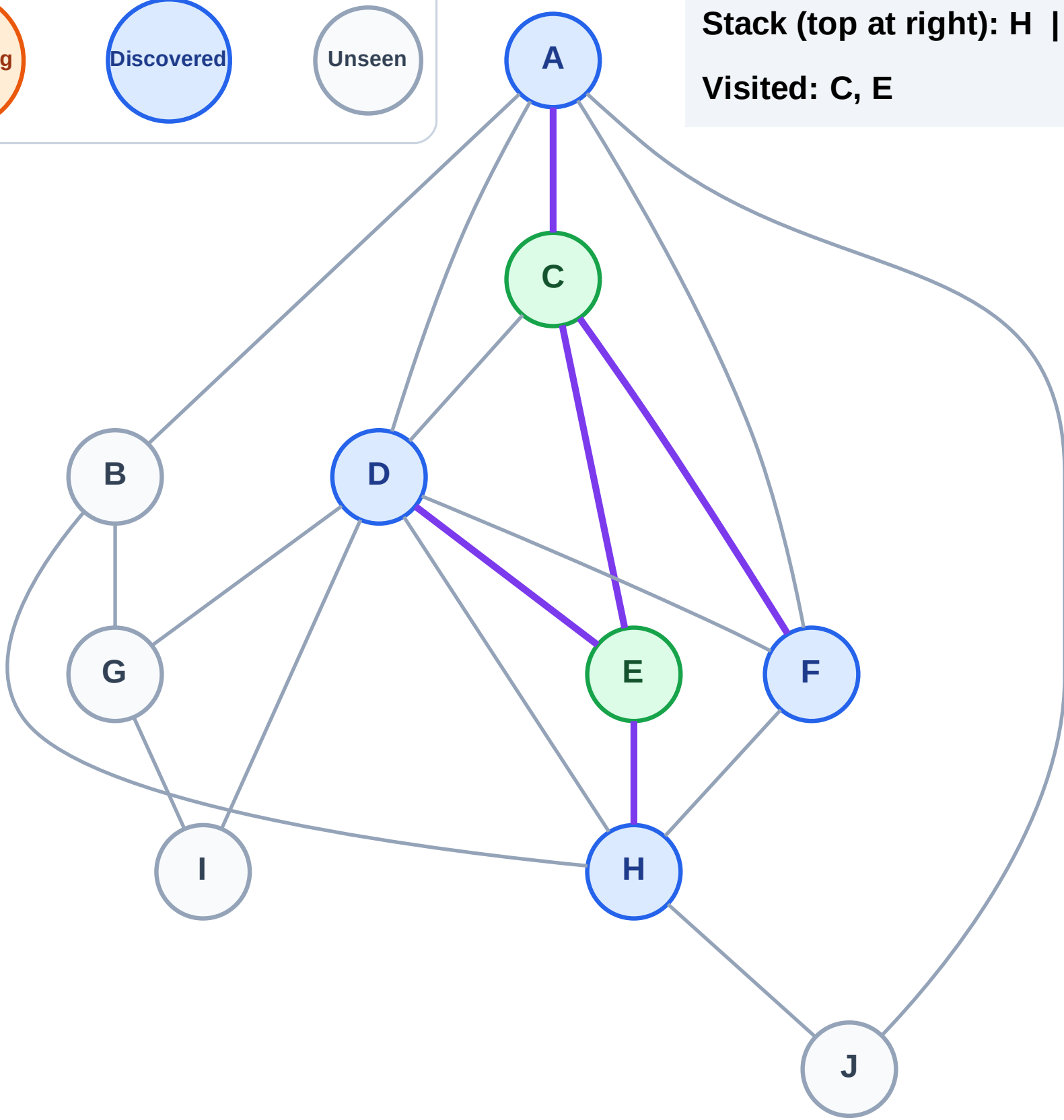
Step 5: Discover F, A from C

## Legend



Stack (top at right): H | D | F | A

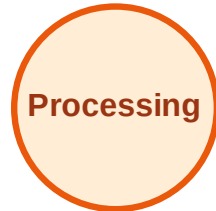
Visited: C, E



# DFS Traversal

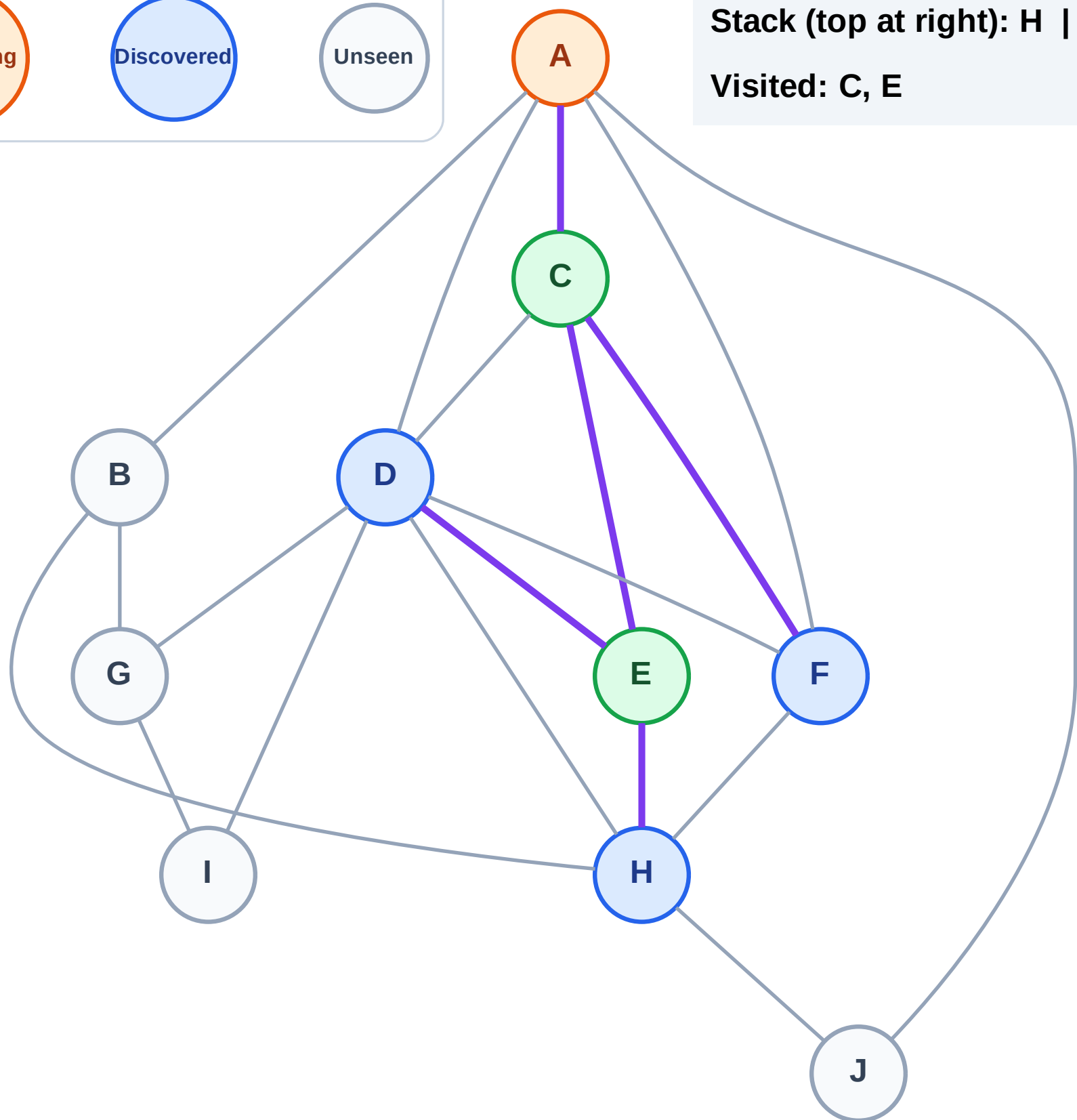
Step 6: Process node A

## Legend



Stack (top at right): H | D | F

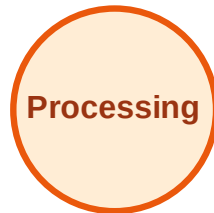
Visited: C, E



# DFS Traversal

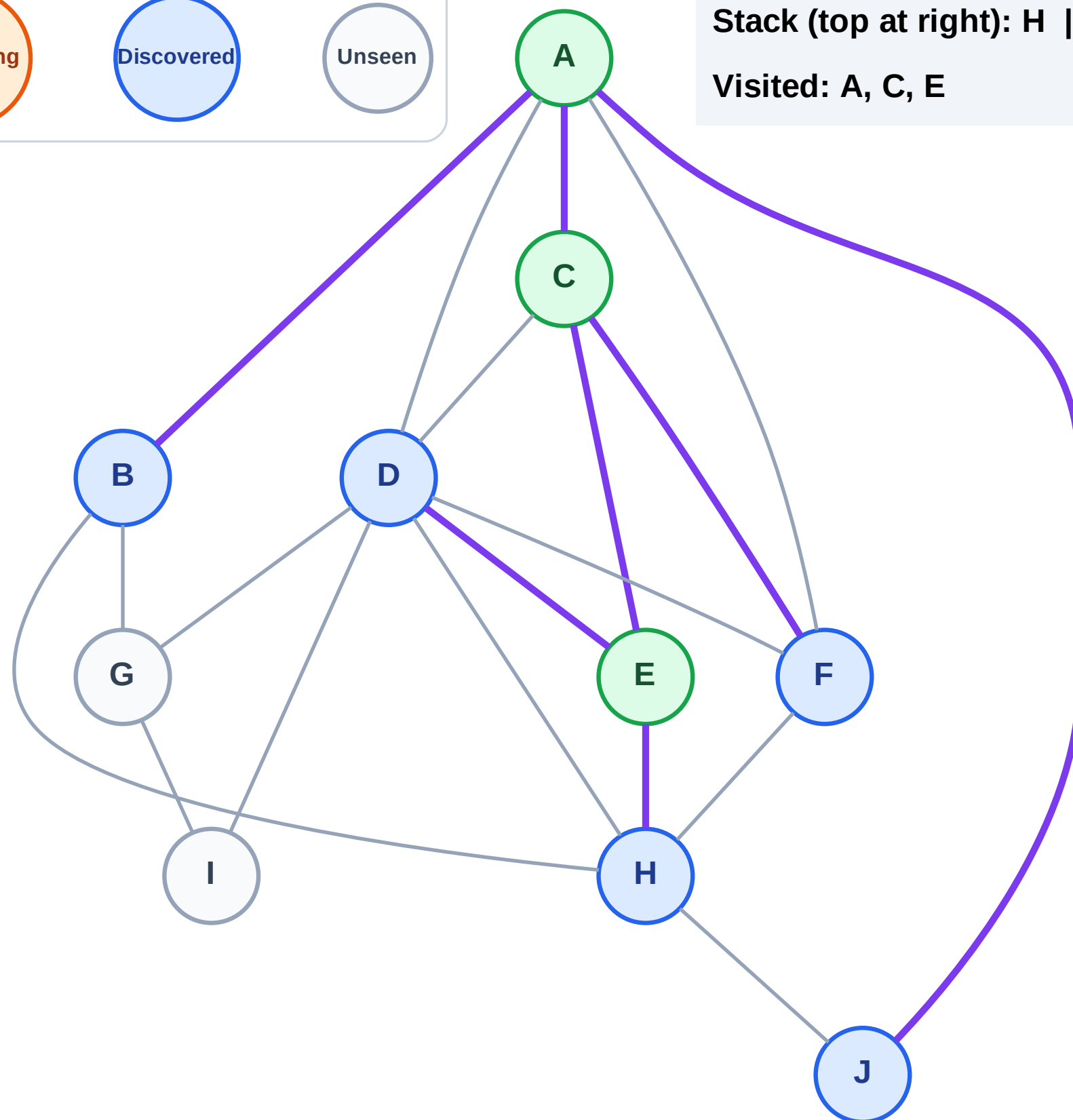
Step 7: Discover J, B from A

## Legend



Stack (top at right): H | D | F | J | B

Visited: A, C, E



# DFS Traversal

Step 8: Process node B

Legend

Complete

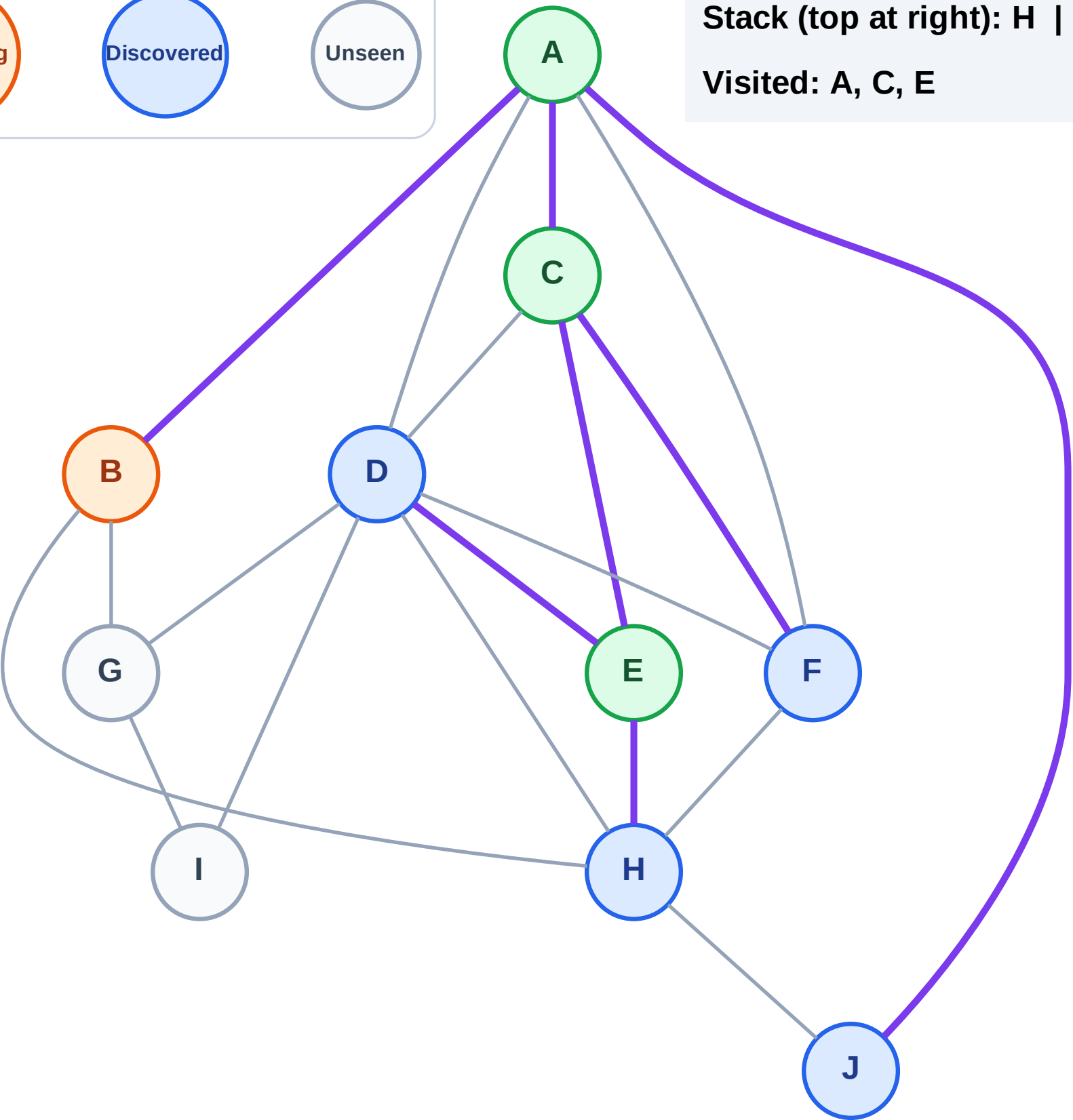
Processing

Discovered

Unseen

Stack (top at right): H | D | F | J

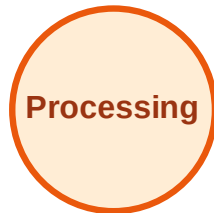
Visited: A, C, E



# DFS Traversal

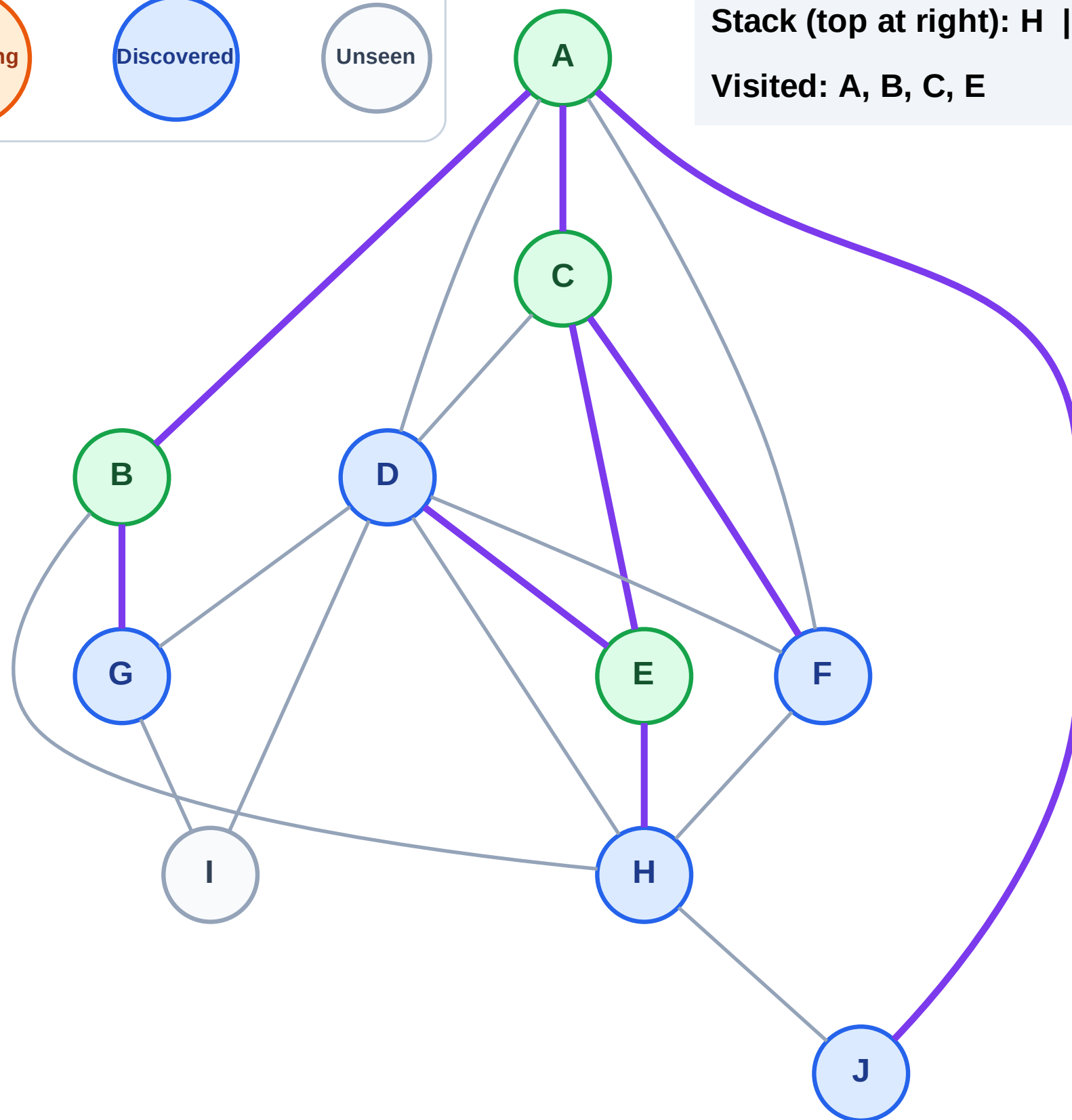
Step 9: Discover G from B

## Legend



Stack (top at right): H | D | F | J | G

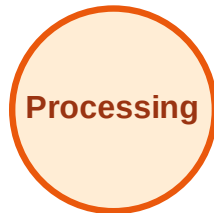
Visited: A, B, C, E



# DFS Traversal

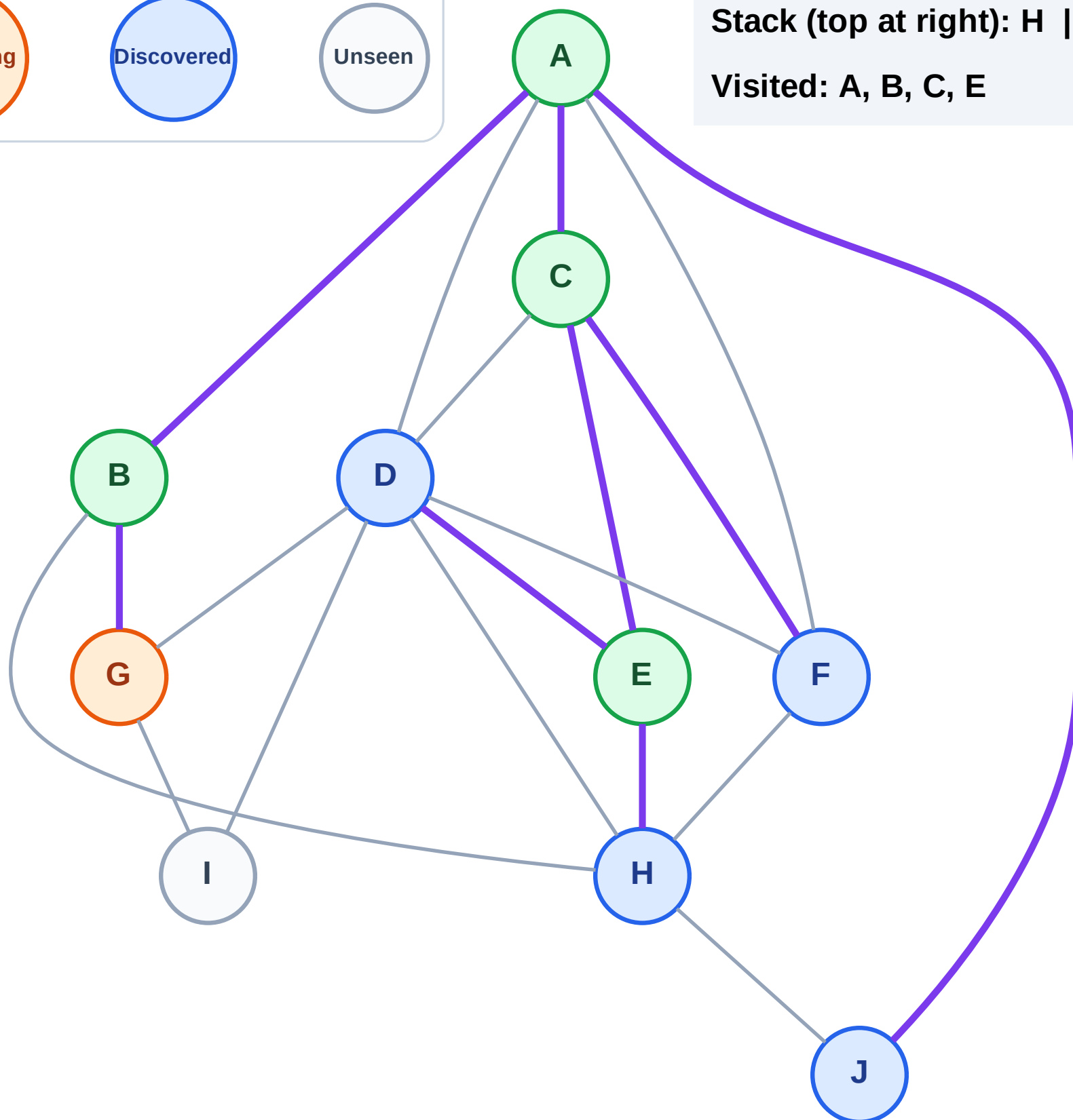
Step 10: Process node G

## Legend



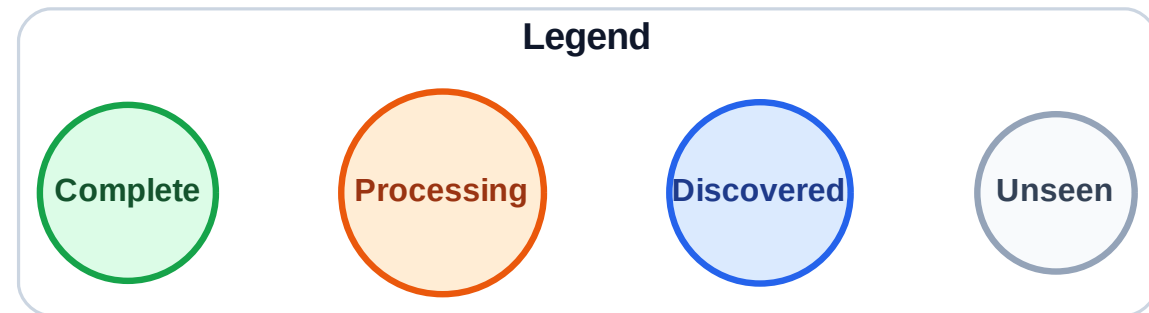
Stack (top at right): H | D | F | J

Visited: A, B, C, E

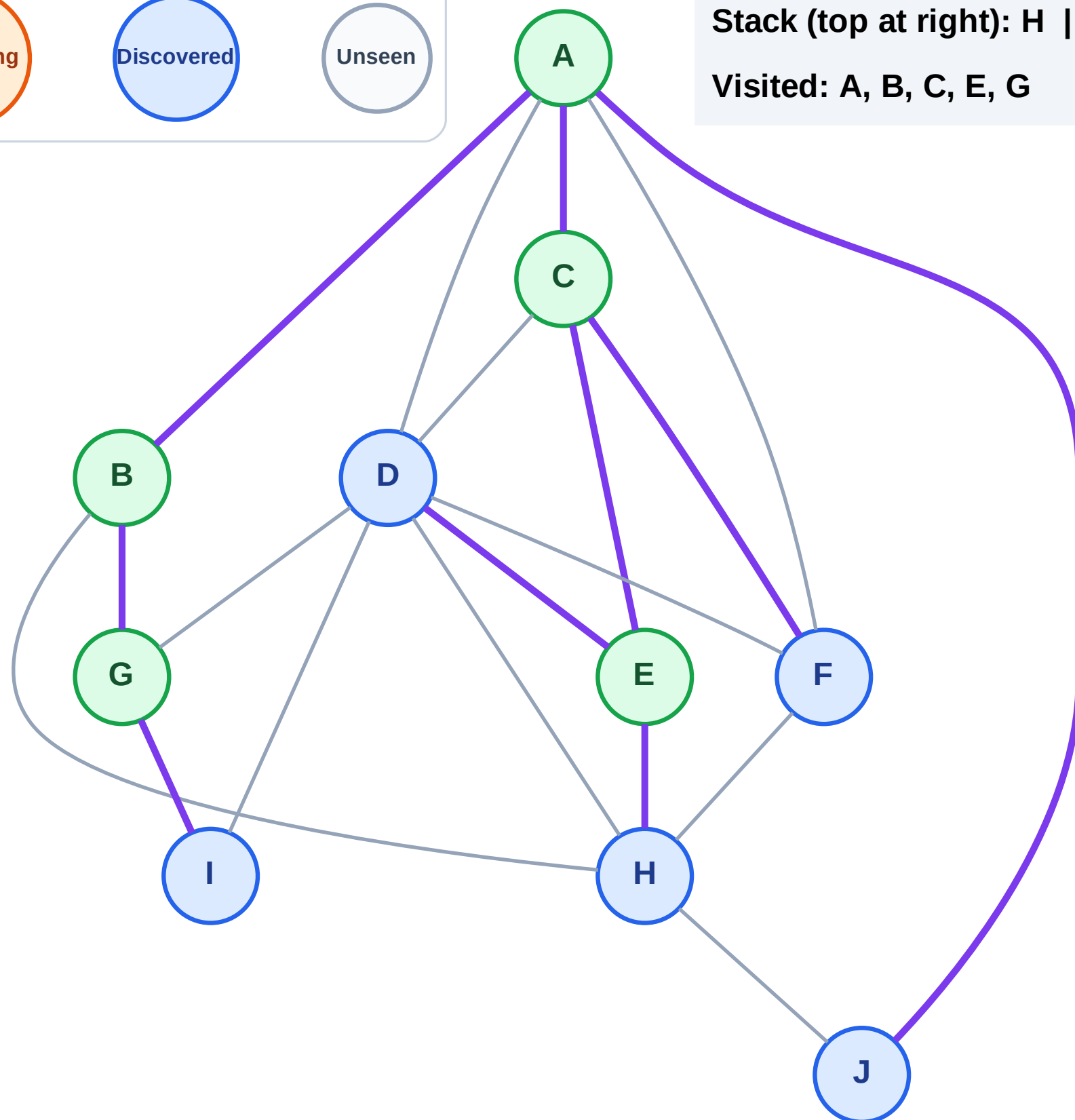


# DFS Traversal

Step 11: Discover I from G



Stack (top at right): H | D | F | J | I  
Visited: A, B, C, E, G



# DFS Traversal

Step 12: Process node I

Legend

Complete

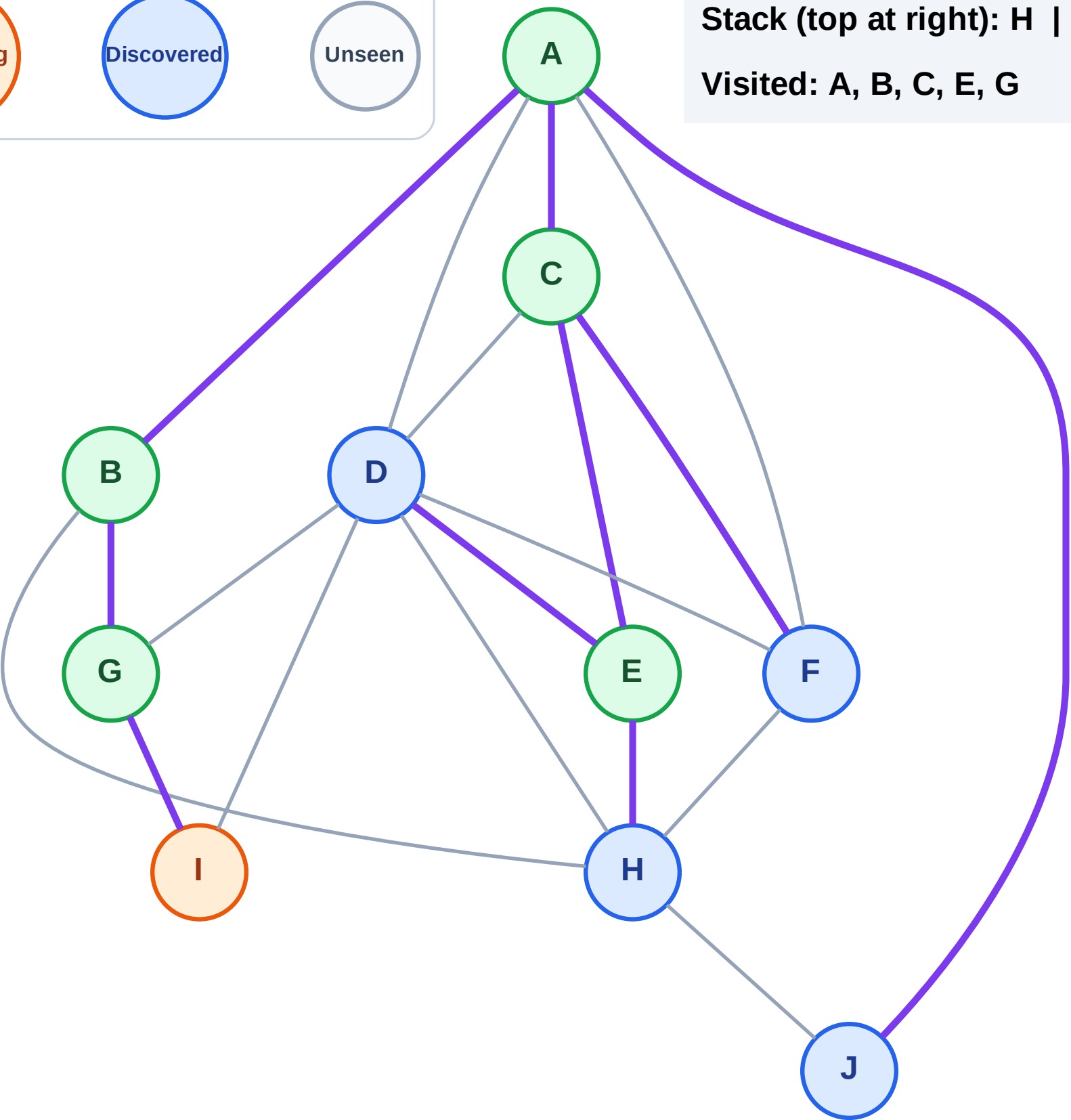
Processing

Discovered

Unseen

Stack (top at right): H | D | F | J

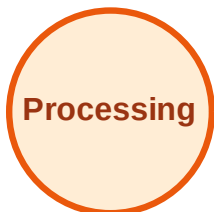
Visited: A, B, C, E, G



# DFS Traversal

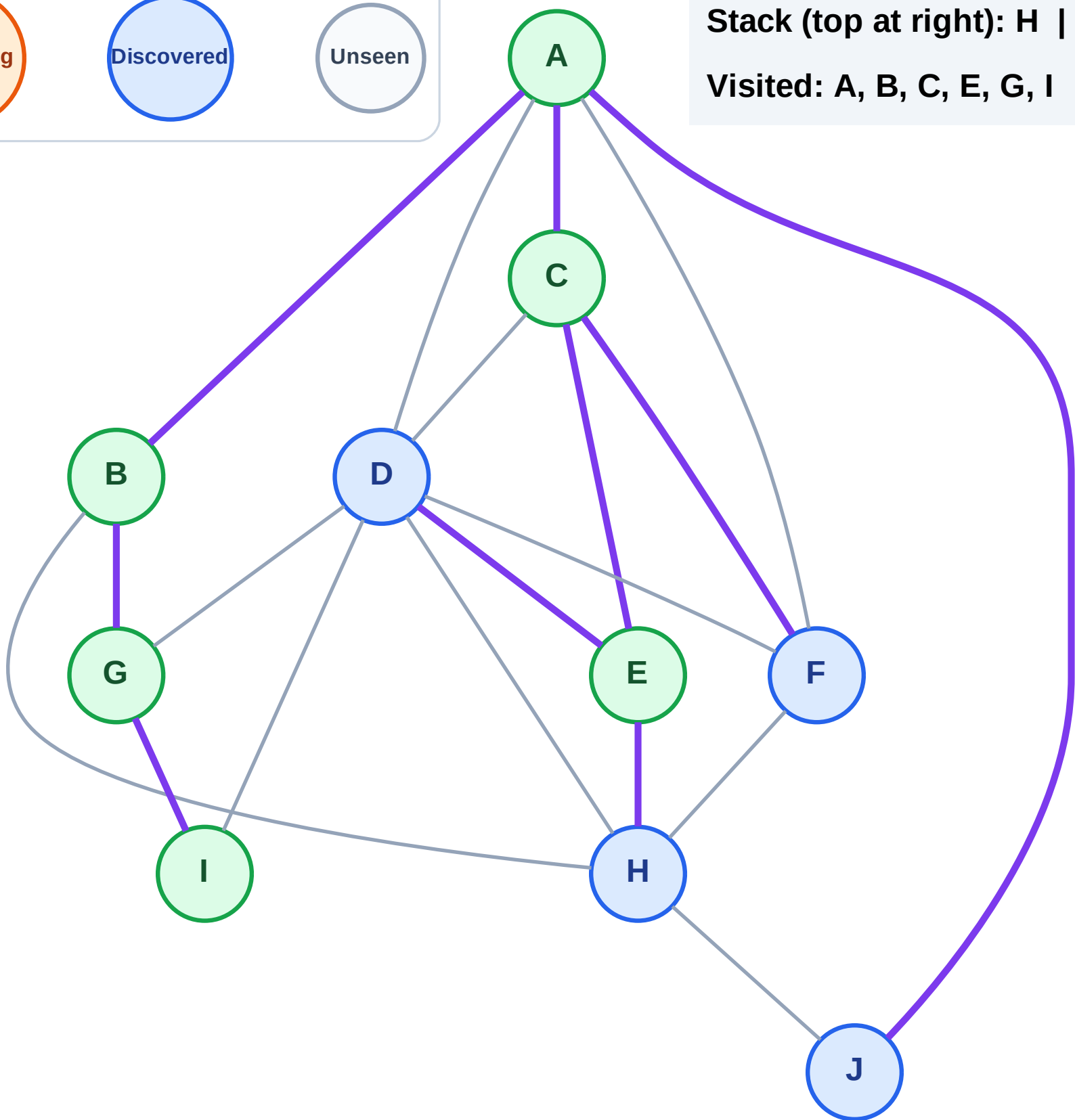
Step 13: No new neighbors from I

## Legend



Stack (top at right): H | D | F | J

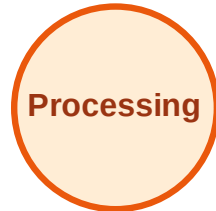
Visited: A, B, C, E, G, I



# DFS Traversal

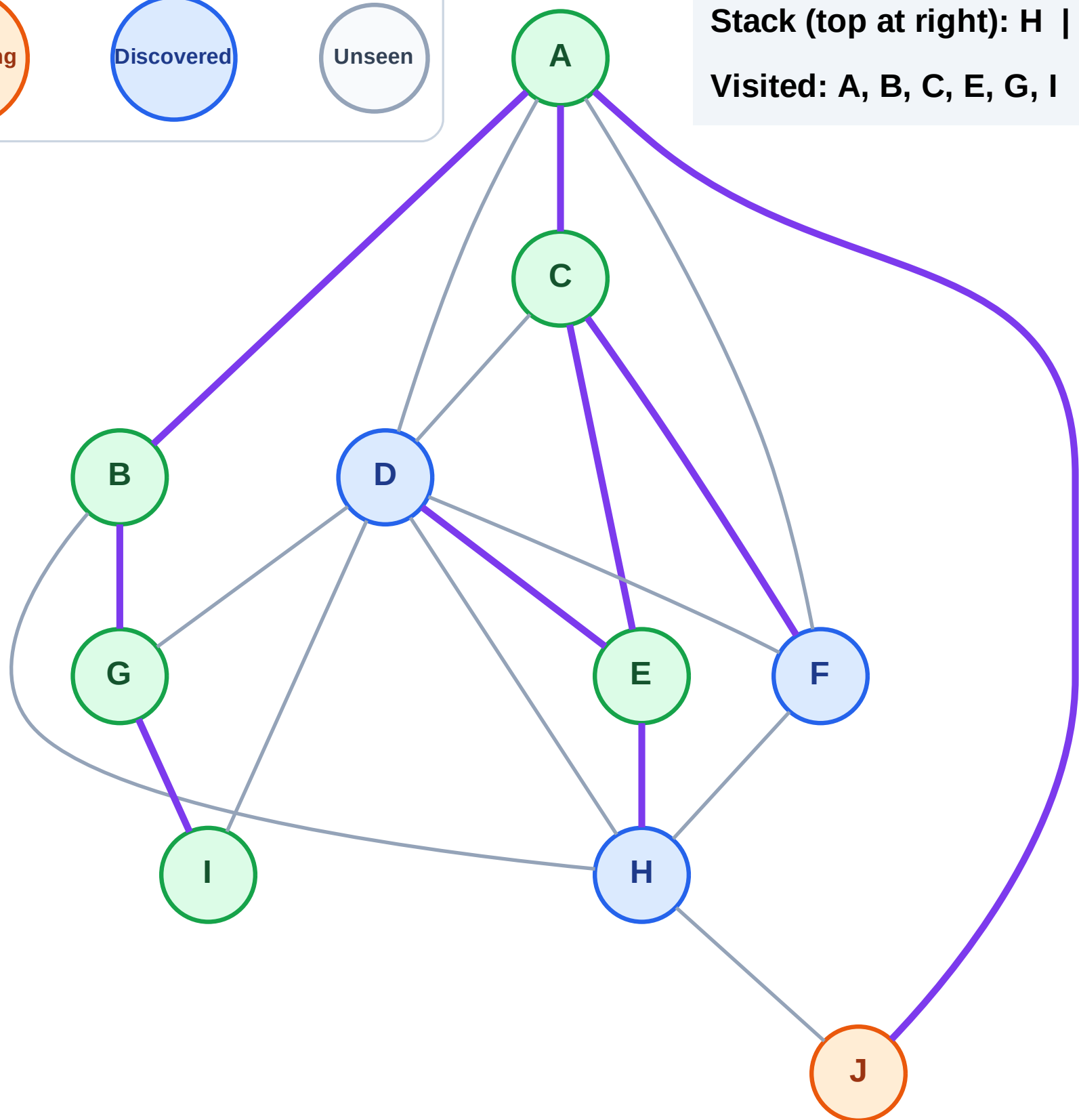
Step 14: Process node J

## Legend



Stack (top at right): H | D | F

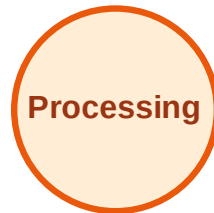
Visited: A, B, C, E, G, I



# DFS Traversal

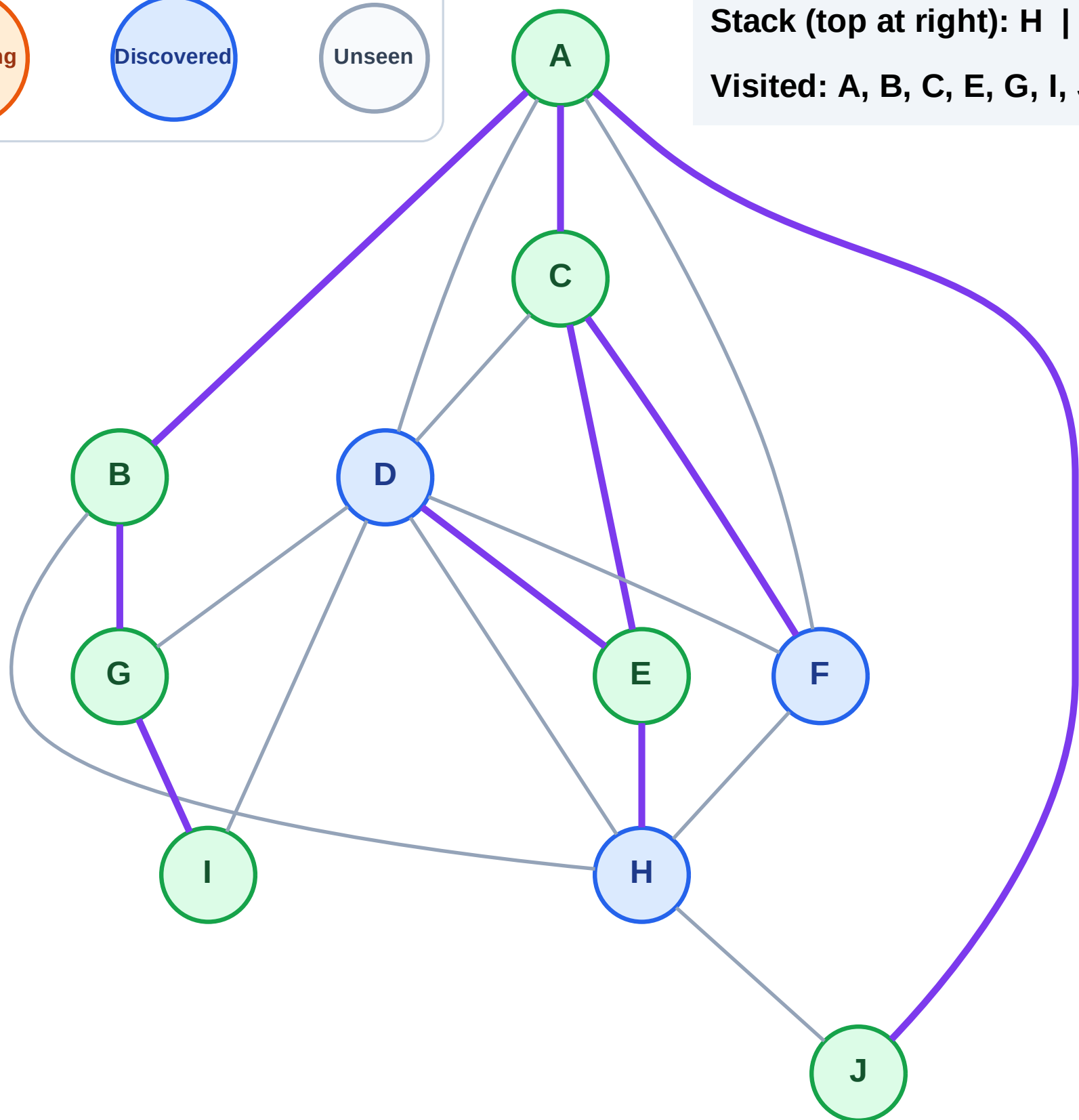
Step 15: No new neighbors from J

## Legend



Stack (top at right): H | D | F

Visited: A, B, C, E, G, I, J



# DFS Traversal

Step 16: Process node F

Legend

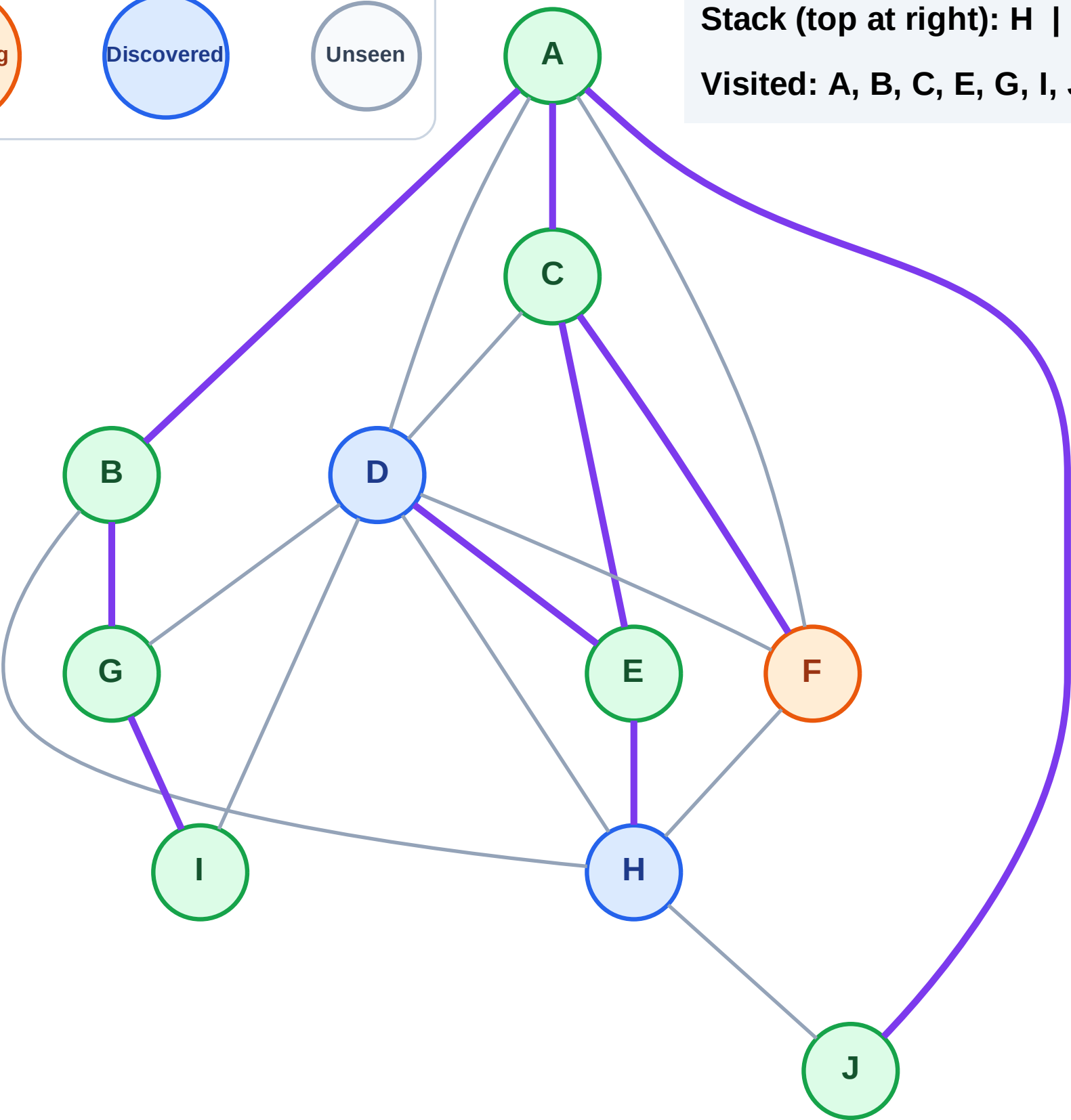
Complete

Processing

Discovered

Unseen

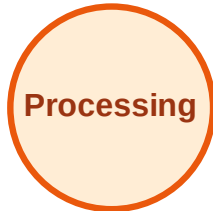
Stack (top at right): H | D  
Visited: A, B, C, E, G, I, J



# DFS Traversal

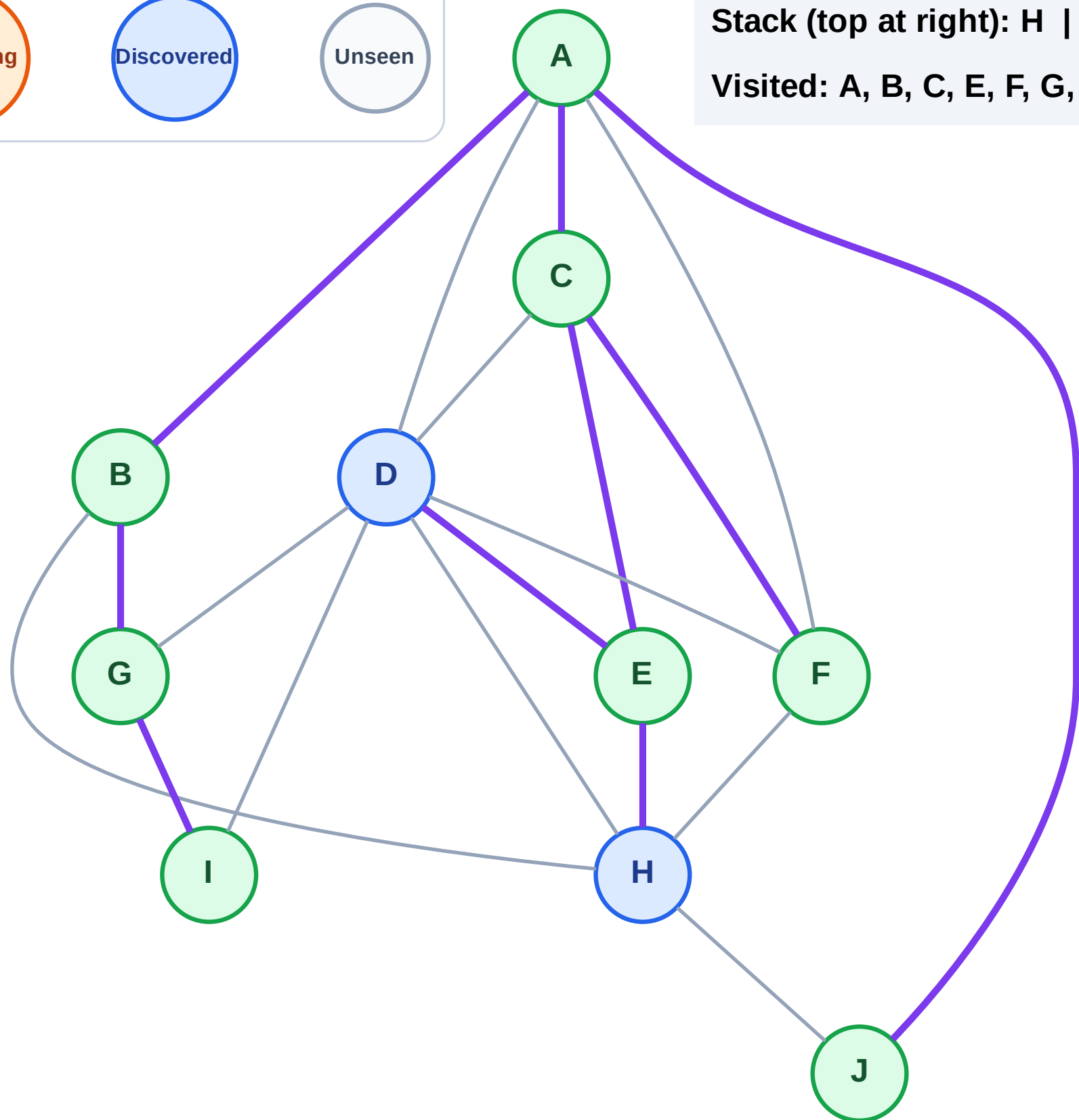
Step 17: No new neighbors from F

## Legend



Stack (top at right): H | D

Visited: A, B, C, E, F, G, I, J



# DFS Traversal

Step 18: Process node D

Legend

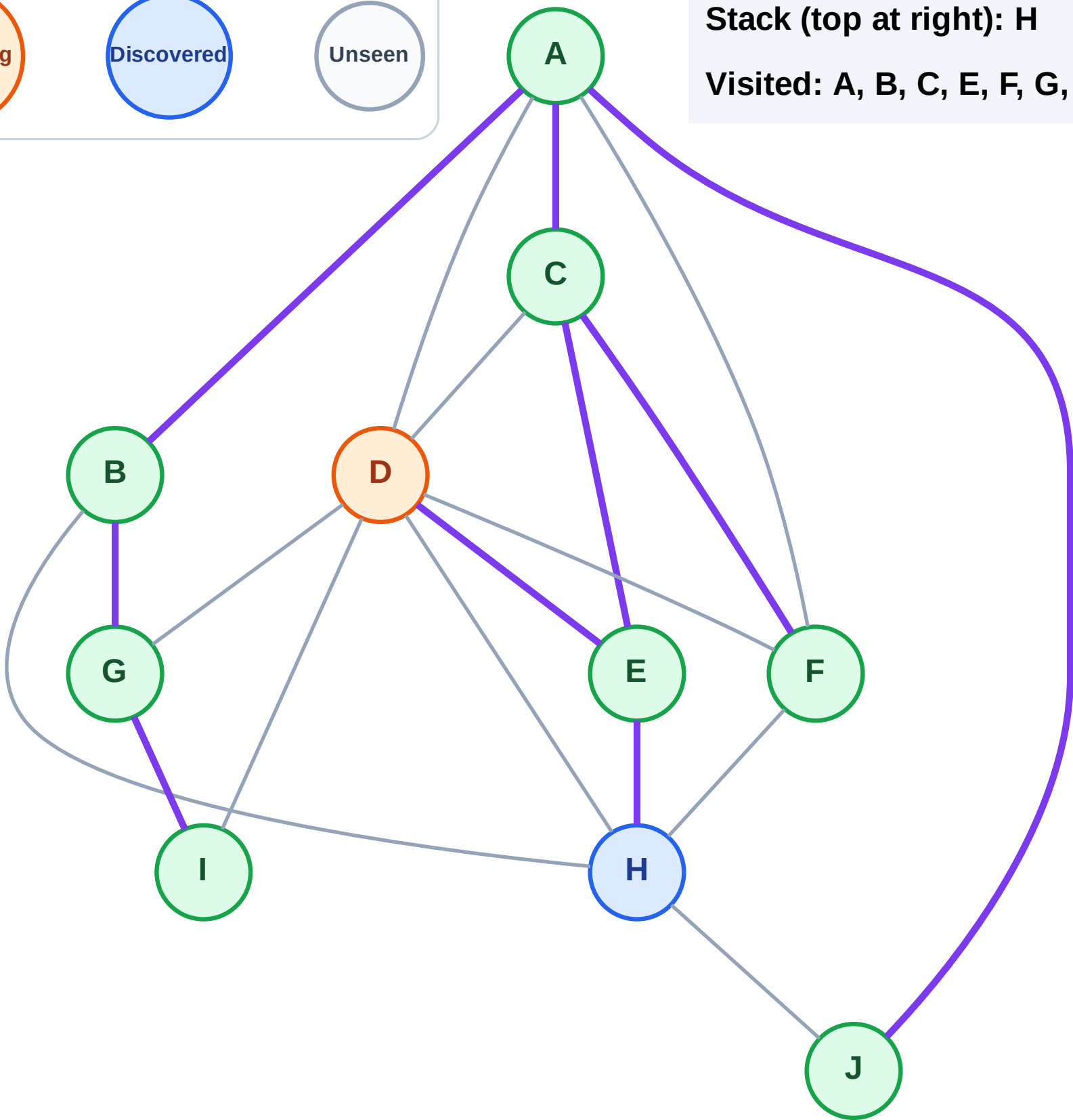
Complete

Processing

Discovered

Unseen

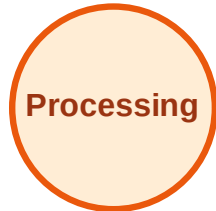
Stack (top at right): H  
Visited: A, B, C, E, F, G, I, J



# DFS Traversal

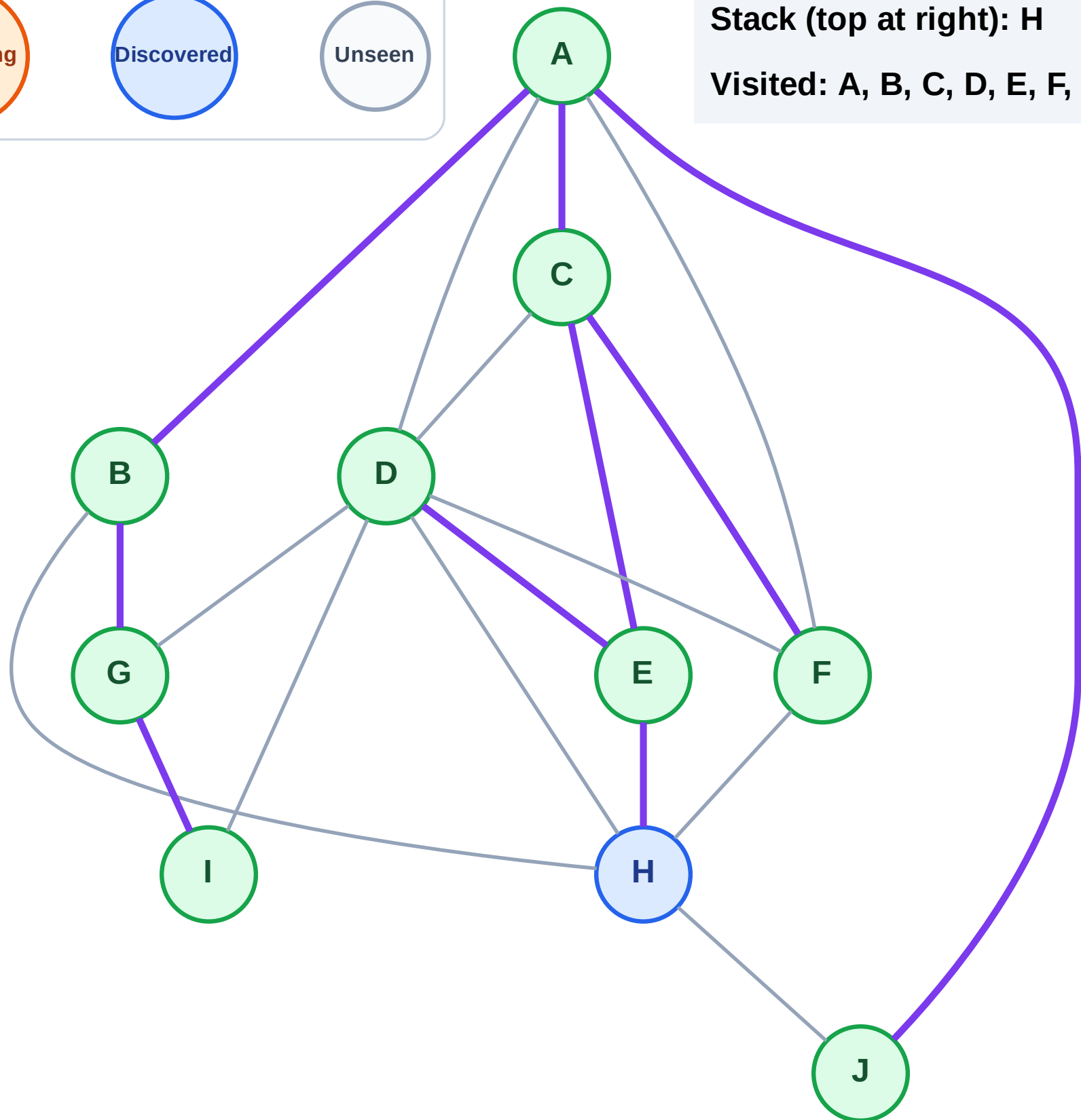
Step 19: No new neighbors from D

## Legend



Stack (top at right): H

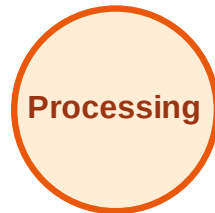
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

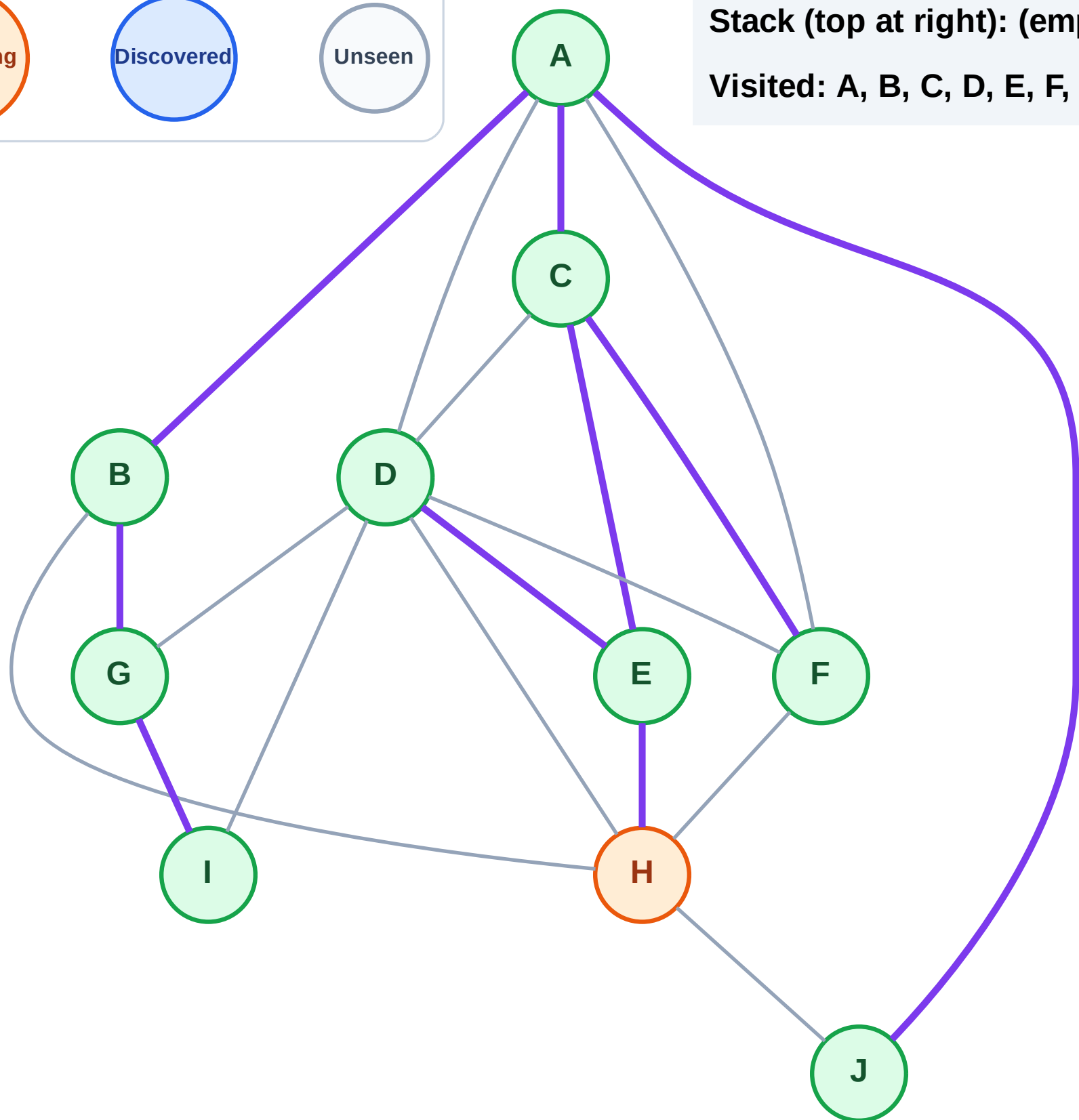
Step 20: Process node H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

